Hyperspectral Real-Time Online Processing Local Anomaly Detection via Multiline Multiband Progressing

Shihui Liu, Meiping Song, *Member, IEEE*, Hui Li, Tingting Yang, Bolun Cui, Xin Li, Jiakang Li, and Dayong Xu

***Abstract*— Anomaly detection, as one of the most critical tasks in hyperspectral image (HSI) processing, has been paid extensive attention in the past decades. The classical work of hyperspectral anomaly detection, sample correlation matrix R-based anomaly detection (R-AD), can achieve desirable detection accuracy, but its processing complexity remains a challenging problem. Real-time anomaly detection methods speed up the procedure through processing the data acquired by grating splitting or acousto-optic tunable filter (AOTF)-based imaging systems in the manner of pixel-by-pixel, line-by-line, or band-by-band. However, in practical industrial scenarios, the multiarray filter spectral imaging system is generally leveraged to avoid high construction and maintenance costs, which acquires multiline multiband (MLMB) data lacking a real-time processing version. In addition, the background of industrial assembly lines is chaotic, and the changes between batches are rapid, both increasing the difficulty of accurate anomaly detection online. To cope with these challenges, a real-time online processing version [real-time multiline multiband R anomaly detection (RTMLMB-RAD)] of R-AD based on MLMB data is proposed here for the first time. Specifically, a multiline multiband correlation matrix (MLMBCM) is designed to update the detector of R-AD recursively with MLMB data acquisition. Since MLMBCM relies only on the previous result and the current data, a large amount of data storage and complex matrix inversion calculation are avoided. At the same time, local anomaly detection mode is adopted to improve the sensitivity of anomalies in different batches and different types of products. The experimental results on six public hyperspectral datasets demonstrate that the proposed RTMLMB-RAD algorithm outperforms other state-of-the-art methods in terms of time cost and detection accuracy. Particularly, the proposed algorithm was tested on the tobacco primary processing line with a detection rate of 87% and a false alarm rate of 10% at a speed of 3 m/s, further verifying the algorithm's feasibility and effectiveness in practical industrial scenarios.**

***Index Terms*— Hyperspectral images (HSIs), local anomaly detection, matrix inversion, multiarray filter spectral imaging system, multiline multiband (MLMB), R-based anomaly detection (R-AD), real-time online processing.**

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I. Introduction

HYPERSPECTRAL imagery provides abundant and detailed spectral information, which can be effectively utilized to distinguish the physical properties of different materials [1], [2], [3], [4], [5], [6]. Anomaly detection, as one of the most critical tasks in hyperspectral image (HSI) processing, has attracted substantial attention and has been widely used in various fields such as agriculture and forestry monitoring, mining exploration, and military defense [7], [8], [9], [10], [11], [12], [13], [14].

In recent years, a wide variety of hyperspectral anomaly detection approaches have emerged. As a pioneering work in this field, Reed–Xiaoli detector (RXD) developed by Reed and Yu [15] is classical and widely used, which assumes that the probability density function of the background (BKG) conforms to multivariate Gaussian distribution. Subsequently, Chang and Chiang [16] proposed R-based anomaly detection (R-AD) based on a correlation matrix, which simplified RXD to some extent. Recently, Chang [17] developed a virtual variable technique that extends constrained energy minimization (CEM) to CEM anomaly detector (CEMAD) in the case of an unknown specific target signature. For RXD only uses original spectral features, resulting in a lack of signal enhancement and noise suppression capabilities, Tao et al. [18] proposed a fractional Fourier entropy (FrFE)-Reed-Xiaoli (RX) method for hyperspectral anomaly detection based on FrFE, which enhanced the distinction between anomaly and BKG by removing noise. In addition to the above statistical theory-based methods, representation-based anomaly detection methods have also been extensively studied in recent years. For example, based on the sparse representation model, Wu et al. [14] proposed an anomaly detection method based on joint sparse representation, presenting local neighborhood pixels around test pixels through joint sparse representation with a BKG dictionary. The machine learning-based method broadens the prospect of hyperspectral anomaly detection; Zhong et al. [19] proposed

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a new representation of BKG anomaly separability with the generative adversarial network (BASGAN) for hyperspectral anomaly detection. By combining model-driven low-rank prior and data-driven automatic encoder, Wang et al. [20] proposed a deep low-rank prior method (DeepLR). Apart from the earlier methods, inspired by the physical characteristics of the HSIs, Li and Zhang [21] developed a hyperspectral anomaly detection method based on image super-resolution and spatial correlation. The above methods use statistics of BKG pixels in the whole hyperspectral imagery, being regarded as global methods and performing well in most cases.

The global methods are not satisfactory with weak anomalies or complicated BKGs. In contrast, local methods, which utilize statistics from neighborhoods, have been extensively investigated recently. For instance, Matteoli et al. [22] established a model to evaluate the impact of signal pollution and its variability on the adaptive detection performance of local anomaly. Based on the low-rank model, Chang et al. [23] proposed a local hyperspectral anomaly detection method based on low-rank and sparse matrix decomposition (LRaSMD) and achieved better detection results. Based on the low-rank and collaborative representation model, Su et al. [9] proposed a hyperspectral anomaly detection model and construction method of the relevant BKG dictionary, which improved the stability of the method. Combining global and local dimensional structures of the BKG together, Wu et al. [24] proposed to employ scalable low-dimensional manifold modeling (SLDMM) to reconstruct the BKG from subsamples of hyperspectral images. Xiang et al. [25] proposed an anomaly detection method based on the local joint subspace process and support vector machine (SVM), to leverage spatial–spectral information and obtain the local subspace score. Although the above global and local anomaly detection algorithms have achieved good results on the hyperspectral images tested, the calculation time is long because of matrix inversion, neighborhood window sliding, or network parameters, and they are not suitable for online real-time detection. It is worth noting that most of the above methods have high computation requirements.

In actual industrial process lines, the anomaly may appear in a short time unpredictably due to the speed movement of transmission media. This requires the system to collect data and complete detection pretty quickly and accurately. Unfortunately, although hyperspectral images benefit much from the high spectral resolution, it also brings about huge data storage and processing complexity. So far, the processing speed has become a bottleneck hindering the application of anomaly detection methods in industrial scenarios [26], [27]. Many efforts have been made to alleviate this obstacle [28], [29], [30]. For example, Chen et al. [31] developed causal processing to perform real-time anomaly detection, where Kalman filters were derived via a recursive causal update equation. In the following year, a real-time CEM (RT-CEM) method was proposed [32], [33] for band-interleaved-pixel/sample (BIP/BIS) and band-interleaved line (BIL) data based on the causal sampling correlation matrix. Subsequently, they put forward a progressive band processing method based on band sequence data for band-by-band anomaly detection (PBP-AD). These methods do not need the complete acquisition of the whole data, so that detection can be realized in a real-time pattern paralleling data processing and data collection [34]. As another example, Díaz et al. [35] proposed a line-by-line fast anomaly detector (LbL-FAD), which was utilized to calculate the orthogonal subspace spanned by selected BKG samples, thus making it easier to detect anomalies. Zhao et al. [36] proposed real-time anomaly detection based on the kernel cooperative representation detector (RT-KCRD), which used a recursive strategy of the kernel covariance matrix to eliminate redundant calculation. A et al. [37] proposed a fast local RX (FLRX) algorithm which effectively utilizes the spectral information and spatial information of hyperspectral images to improve detection accuracy and reduce the running time. Hupel and Stütz [38] developed a local point density (LPD) for near real-time camouflage detection method in multispectral imagery. It can be seen that the data acquisition of the above methods is pixel-by-pixel, line-by-line, or band-by-band. Although good detection results can be obtained, there is a problem with the high cost of data acquisition equipment. To reduce deployment and maintenance costs, multiarray filter spectral imaging cameras are preferred instead of grating splitting or acousto-optic tunable filter (AOTF)-based imaging cameras when the number of necessary cameras is great in industrial production. Therefore, how to deal with multiline multiband (MLMB) data in a real-time way is of significant importance.

To the best of authors' knowledge, the real-time online processing anomaly detection method based on MLMB data has not been investigated in the literature. Motivated by this, we propose a real-time online processing local anomaly detection algorithm (RTMLMB-RAD) based on R-AD for MLMB data. The major contributions of this work are summarized as follows.

1) We have proposed a real-time online processing anomaly detection algorithm for MLMB data obtained by a multiarray filter spectral imaging camera. This provides support for low-cost and real-time processing of industrial applications because it does not need to wait for the complete data, store it, and then calculate it, which will take a long time.
2) For the MLMB data acquisition mode, we have designed multiline multiband correlation matrix (MLMBCM) to replace the correlation matrix in R-AD. Because MLMBCM only consists of the result of the previous state and is updated by vector calculation with the data being processed, it avoids huge data storage and complex matrix inversion of the whole data.
3) We have proposed a local R-AD real-time version based on MLMBCM, which uses local information to detect anomalies in complex BKGs and improves the response degree to different batches of different types of products.

The rest of this article is organized as follows. In Section II, the hyperspectral real-time online processing local anomaly detection algorithm is proposed. Section III gives the corresponding computation and time complexity analysis. In Section IV, we provide extensive experimental results on several datasets to demonstrate the performance of the proposed algorithm. Section V concludes this article.

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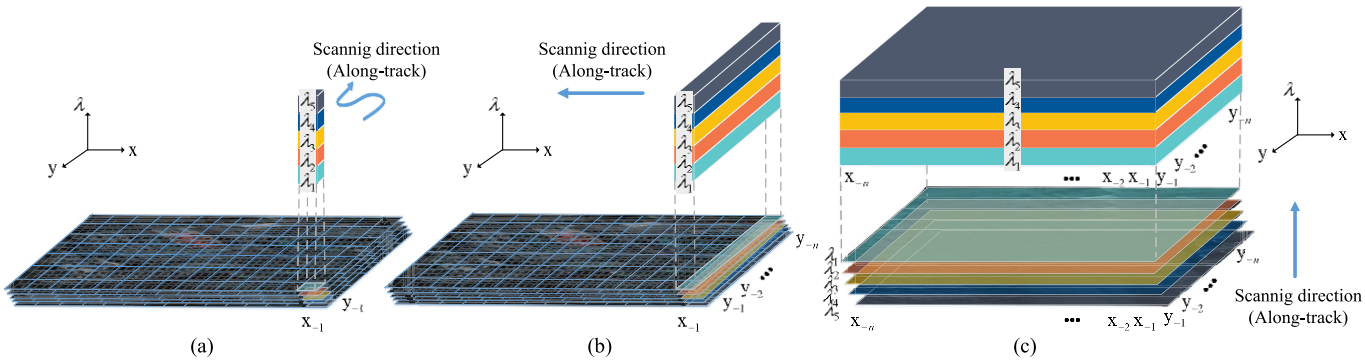


Fig. 1. Diagram of three different data acquisition formats: (a) BIP/BIS; (b) BIL; and (c) BSQ.

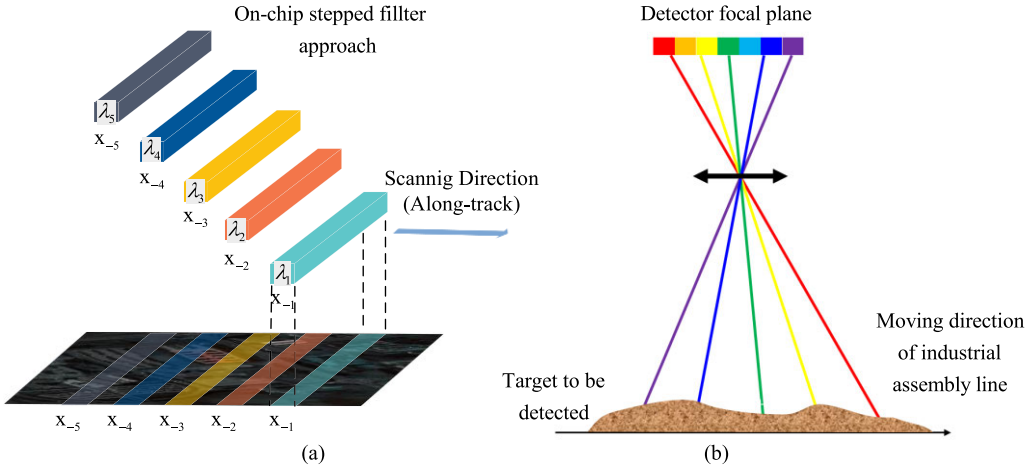


Fig. 2. Schematic of filter light splitting array pushbroom imaging principle: (a) filter pushbroom imaging principle and (b) filter pushbroom geometry.

II. Methodology

In this section, we first introduce the concept of MLMB data type. Then, the MLMBCM is described and RTMLMB-RAD using MLMBCM is provided in detail.

A. MLMB Data Type

The data acquisition format of the spectral imaging camera determines the real-time data processing method. Currently, there are three common data acquisition formats for hyperspectral imaging sensors, which are BIP/BIS, BIL, and band sequential (BSQ) formats, respectively. Among them, BIP/BIS collects data pixel-by-pixel with full band information, as shown in Fig. 1(a). BIL collects data line-by-line with the information of the full band information, as shown in Fig. 1(b). BSQ collects information for the whole data band-by-band, as shown in Fig. 1(c). Compared with the previous two kinds of collection methods, it has spatial information, but there are spectral deformation problems. The above three kinds of data are collected by spectral camera based on grating splitting or AOTF, which is complicated and makes the camera large and expensive, not suitable for lightweight and low-cost industrial scenarios.

Filter film spectral imaging cameras can be developed by adding the filter films into the imaging system [38], [39], which has the advantages of simple structure, high stability, and low cost. Therefore, to collect different band information at the same time, an array filter is usually used for the spectral imaging camera, with different bands for different lines. Multiple independent filters were sequentially spliced into a filter array and placed in front of the imaging units, i.e., multiarray filter spectral imaging, as shown in Fig. 2(a), to obtain data of different spectral wavelength and position of the target, as shown in Fig. 2(b). Along with the proceeding of the transmission media, data containing spectral and spatial information of the whole scene is completed.

In an imaging system using a multiarray filter spectral imaging camera, each imaging of the detector is information in multiple lines and multiple bands. To more clearly illustrate the process of data acquisition, the schematic of data extraction for different lines and different bands is shown in Fig. 3, where the transmission belt moves to the right. Among them, different colors represent detectors with different filters.

As can be seen from Fig. 3, Fig. 3(a) is the state before reading data. According to the moving direction, the gradual acquisition process of different bands of data is realized, as shown in Fig. 3(b)–(g). To more intuitively reflect the process of acquiring data in Fig. 3, the 2-D data of the hyperspectral analog image obtained is shown in Fig. 4. The first number with color in Fig. 4 represents the number of lines corresponding to the original data, and the second number in Fig. 4 corresponds to the band of the original data. It can be seen that data of the same color constitutes a new data line.

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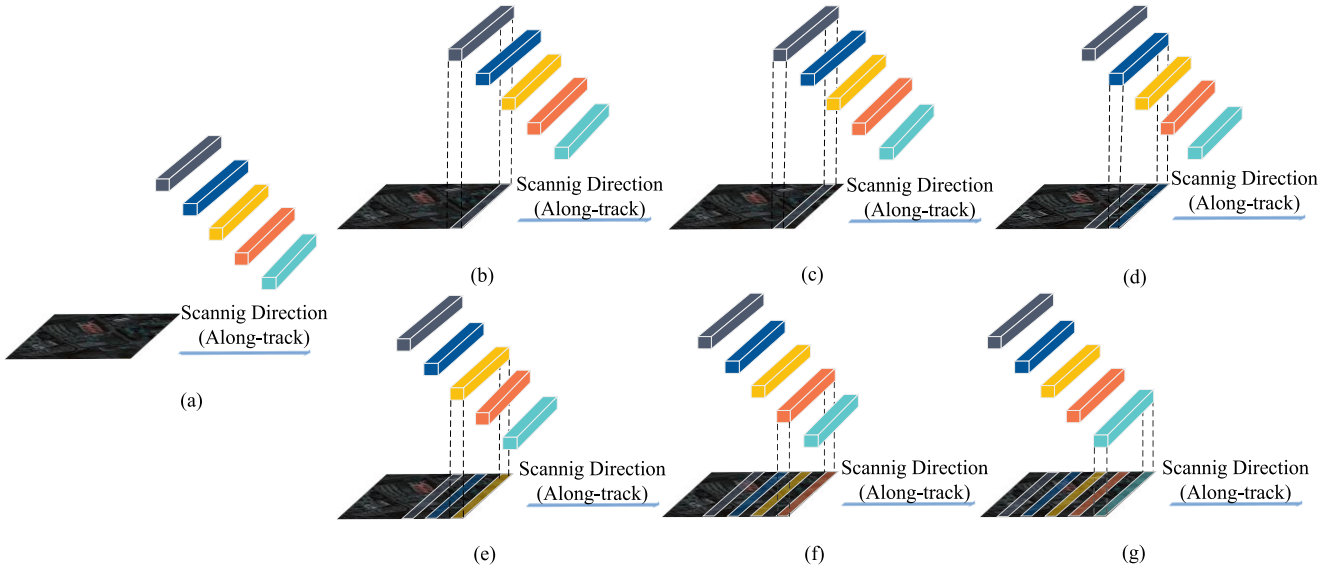


Fig. 3. Example of data extraction in different lines and different bands. (a) Initial state. (b)–(g) Data extraction process diagrams.

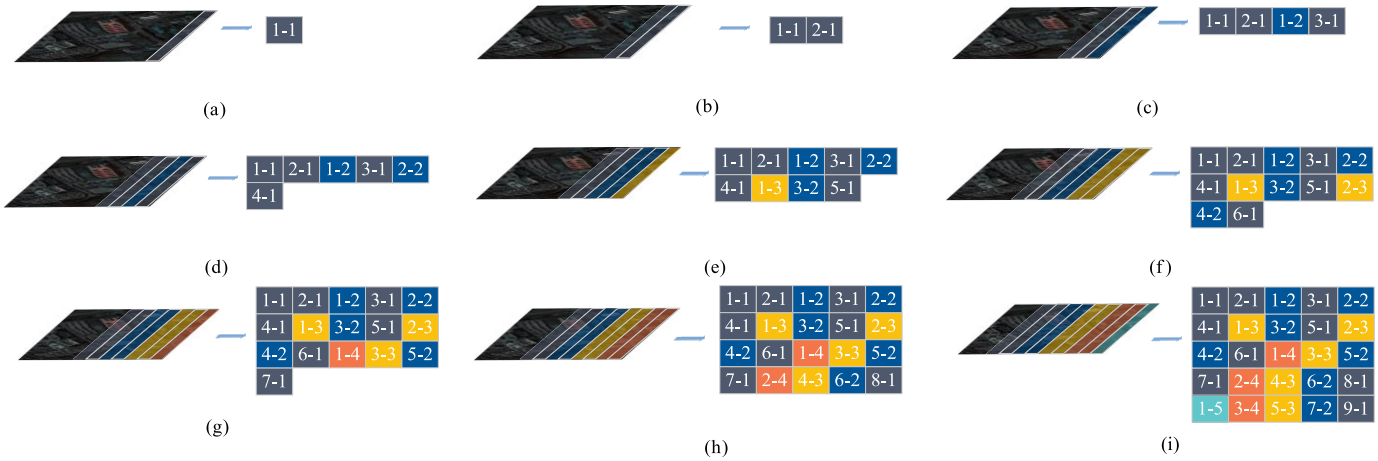


Fig. 4. Bands and lines of the 2-D data corresponding to the original data in the hyperspectral image. (a)–(i) 2-D data extraction process diagrams.

According to the size of the sample data, there are five bands in a line, and each band contains the corresponding pixel value, as shown in Fig. 4. It can be seen that the simulated 2-D data correspond to the number of lines and bands of the original data. The data in Fig. 4(a)–(h) have one or more different bands in the same line, while the data after Fig. 4(i) have five different bands in different lines. It can be seen from Fig. 4 that the hyperspectral simulation image is irregular 2-D data. In other words, once imaging 2-D data contain different bands in different lines. For such data, the correlation matrix cannot be directly calculated, and anomaly detection cannot be implemented. Therefore, to realize real-time online processing anomaly detection on this data type, it needs to be processed and converted into relatively regular 2-D data, as shown in Fig. 5.

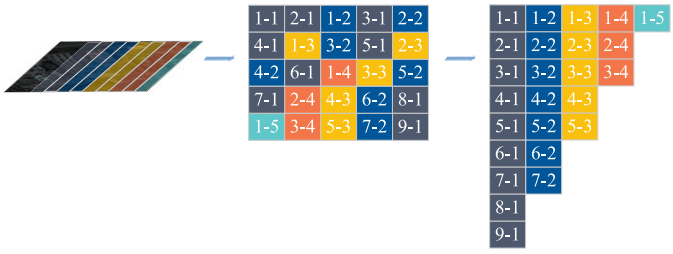


Fig. 5. Bands and number of lines corresponding to the original data of 2-D regular data in hyperspectral simulation image.

B. Multiline Multiband Correlation Matrix

In this section, given the mechanism of a spectral imaging system using a multiarray filter spectral imaging camera, the MLMBCM and the recursive update equation of its inverse matrix are designed. When obtaining the band information of each line, the algorithm does not need to recalculate the all-data correlation matrix and its inverse matrix, and only uses the current state and new information for the recursive solution.

Assume that $\mathbf{x}^m(l) = (r^m_{1l}, r^m_{2l}, \ldots, r^m_{(M-1)l}, r^m_{Ml})^{\mathrm{T}}$ is all pixel reflectance of the lth band in the mth line of 2-D data $\mathbf{X}$ simulating hyperspectral image and M is the total number of pixels in the original data line. The number of bands obtained

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from the data is n, and the value range of l is $1-n$. $\mathbf{r}_l^m = (r_1^m, r_2^m, \ldots, r_l^m)^{\mathrm{T}}$ is the data of the first l bands of a specific pixel in mth line of 2-D data $\mathbf{X}$ simulating hyperspectral image, and r_l^m is the reflectance of the lth band of a specific pixel in mth line of 2-D data simulating hyperspectral image $\mathbf{X}$. If $\mathbf{X}_n^m$ is all the data of n bands in the mth line of the simulated 2-D hyperspectral image $\mathbf{X}$, the $\mathbf{X}_n^m$ data size is $n * M$, and the specific expression of $\mathbf{X}_n^m = (\mathbf{x}^m(1), \mathbf{x}^m(2), \ldots, \mathbf{x}^m(n))^{\mathrm{T}} = [\mathbf{r}_n^1, \mathbf{r}_n^2, \ldots, \mathbf{r}_n^m]$ is given as follows:

$$\begin{aligned}\mathbf{X}_n^m &= \left(\mathbf{x}^m(1), \mathbf{x}^m(2), \ldots, \mathbf{x}^m(n)\right)^{\mathrm{T}} \\ &= \begin{bmatrix} r_{11}^m & \cdots & r_{(M-1)1}^m & r_{M1}^m \\ \vdots & \ddots & \vdots & \vdots \\ r_{1(n-1)}^m & \cdots & r_{(M-1)(n-1)}^m & r_{M(n-1)}^m \\ r_{1n}^m & \cdots & r_{(M-1)n}^m & r_{Mn}^m \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{X}_{n-1}^m \\ \mathbf{x}^m(n)^{\mathrm{T}} \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{X}_{n-1}^m \\ \mathbf{U} \end{bmatrix} \end{aligned} \tag{1}$$

where $\mathbf{X}_{n-1}^m$ is as shown in (2), $\mathbf{U} = (\mathbf{x}^m(n))^{\mathrm{T}}$ size is $1 * M$, and $\mathbf{U}$ represents all the pixels of the nth band in mth line of the simulated hyperspectral image $\mathbf{X}$ as shown in (3)

$$\begin{aligned}\mathbf{X}_{n-1}^m &= \left(\mathbf{x}^m(1), \mathbf{x}^m(2), \ldots, \mathbf{x}^m(n-1)\right)^{\mathrm{T}} \\ &= \begin{bmatrix} r_{11}^m & \cdots & r_{(M-1)1}^m & r_{M1}^m \\ \vdots & \ddots & \vdots & \vdots \\ r_{1(n-2)}^m & \cdots & r_{(M-1)(n-2)}^m & r_{M(n-2)}^m \\ r_{1(n-1)}^m & \cdots & r_{(M-1)(n-1)}^m & r_{M(n-1)}^m \end{bmatrix} \end{aligned} \tag{2}$$

$$\mathbf{U} = \begin{pmatrix} r_{1n}^m & \cdots & r_{(M-1)n}^m & r_{Mn}^m \end{pmatrix}. \tag{3}$$

It is worth noting that when $n = 1$, $\mathbf{X}_{n-1}^m$ in (1) is the n bands in the $m-1$th line of $\mathbf{X}$, that is, $\mathbf{X}_{n-1}^m = \mathbf{X}_n^{m-1}$. The n bands of the previous $m-1$ line are used to make better use of the local information of the previous line. In this article, different calculation ways should be used in different situations. Specifically, according to the n value, the MLMBCM is divided into two ways: one is between different lines, and the other is the same line.

1) Multiline Multiband Correlation Matrix Between Different Lines: The multiline multiband correlation matrix between different lines (MLMBCM-D) is defined as all the pixels in the first n bands of the hyperspectral image in the m line, marked as $\mathbf{R}_n^m$. Its expression is shown in the following equation:

$$\begin{aligned}\mathbf{R}_n^m &= \frac{1}{N}\mathbf{X}_n^m\left(\mathbf{X}_n^m\right)^{\mathrm{T}} = \frac{1}{N}\begin{bmatrix} \mathbf{X}_n^{m-1} \\ \mathbf{U} \end{bmatrix}\begin{bmatrix} \left(\mathbf{X}_n^{m-1}\right)^{\mathrm{T}} & \mathbf{U}^{\mathrm{T}} \end{bmatrix} \\ &= \frac{1}{N}\begin{bmatrix} \mathbf{X}_n^{m-1}\left(\mathbf{X}_n^{m-1}\right)^{\mathrm{T}} & \mathbf{X}_n^{m-1}\mathbf{U}^{\mathrm{T}} \\ \mathbf{U}\left(\mathbf{X}_n^{m-1}\right)^{\mathrm{T}} & \mathbf{U}\mathbf{U}^{\mathrm{T}} \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{N}\mathbf{X}_n^{m-1}\left(\mathbf{X}_n^{m-1}\right)^{\mathrm{T}} & \frac{1}{N}\mathbf{X}_n^{m-1}\mathbf{U}^{\mathrm{T}} \\ \frac{1}{N}\mathbf{U}\left(\mathbf{X}_n^{m-1}\right)^{\mathrm{T}} & \frac{1}{N}\mathbf{U}\mathbf{U}^{\mathrm{T}} \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{R}_n^{m-1} & \frac{1}{N}\mathbf{X}_n^{m-1}\mathbf{U}^{\mathrm{T}} \\ \frac{1}{N}\mathbf{U}\left(\mathbf{X}_n^{m-1}\right)^{\mathrm{T}} & \frac{1}{N}\mathbf{U}\mathbf{U}^{\mathrm{T}} \end{bmatrix}. \end{aligned} \tag{4}$$

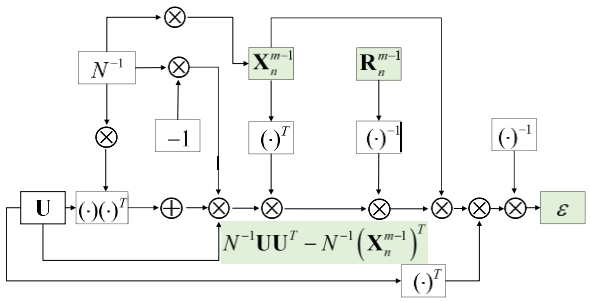


Fig. 6. Schematic of ε in (8).

Anomaly detection can be achieved by inverting the MLMBCM $\mathbf{R}_n^m$. To obtain the recursive update of $(\mathbf{R}_n^m)^{-1}$ from the recursive solution of $(\mathbf{R}_n^{m-1})^{-1}$, the block matrix inversion formula is used for the recursive update solution

$$\begin{aligned}&\begin{bmatrix} \mathbf{A}\mathbf{B} \\ \mathbf{C}\mathbf{D} \end{bmatrix}^{-1} \\ &= \begin{bmatrix} \mathbf{A}^{-1}+\mathbf{A}^{-1}\mathbf{B}(\mathbf{D}-\mathbf{C}\mathbf{A}^{-1}\mathbf{B})^{-1}\mathbf{C}\mathbf{A}^{-1} & -\mathbf{A}^{-1}\mathbf{B}(\mathbf{D}-\mathbf{C}\mathbf{A}^{-1}\mathbf{B})^{-1} \\ -(\mathbf{D}-\mathbf{C}\mathbf{A}^{-1}\mathbf{B})^{-1}\mathbf{C}\mathbf{A}^{-1} & (\mathbf{D}-\mathbf{C}\mathbf{A}^{-1}\mathbf{B})^{-1} \end{bmatrix}. \end{aligned} \tag{5}$$

The following equation can be obtained from (4):

$$\left(\mathbf{R}_n^m\right)^{-1} = \begin{bmatrix} \mathbf{R}_n^{m-1} & \frac{1}{N}\mathbf{X}_n^{m-1}\mathbf{U}^{\mathrm{T}} \\ \frac{1}{N}\mathbf{U}\left(\mathbf{X}_n^{m-1}\right)^{\mathrm{T}} & \frac{1}{N}\mathbf{U}\mathbf{U}^{\mathrm{T}} \end{bmatrix}. \tag{6}$$

Refer to the block matrix of (5) to calculate the inverse equation. Let $\mathbf{A} = \mathbf{R}_n^{m-1}$, $\mathbf{B} = (1/N)\mathbf{X}_n^{m-1}\mathbf{U}^{\mathrm{T}}$, $\mathbf{C} = (1/N)\mathbf{U}(\mathbf{X}_n^{m-1})^{\mathrm{T}}$, and $\mathbf{D} = (1/N)\mathbf{U}\mathbf{U}^{\mathrm{T}}$, then the recursive update of (6) can be obtained by the following equation:

$$\begin{aligned}&\left(\mathbf{R}_n^m\right)^{-1} \\ &= \begin{bmatrix} \mathbf{R}_n^{m-1} & \frac{1}{N}\mathbf{X}_n^{m-1}\mathbf{U}^{\mathrm{T}} \\ \frac{1}{N}\mathbf{U}\left(\mathbf{X}_n^{m-1}\right)^{\mathrm{T}} & \frac{1}{N}\mathbf{U}\mathbf{U}^{\mathrm{T}} \end{bmatrix}^{-1} \\ &= \begin{bmatrix} \left(\mathbf{R}_n^{m-1}\right)^{-1}+\frac{1}{N}\left(\mathbf{R}_n^{m-1}\right)^{-1}\mathbf{X}_n^{m-1}\mathbf{U}^{\mathrm{T}}\alpha\beta & -\frac{1}{N}\left(\mathbf{R}_n^{m-1}\right)^{-1}\mathbf{X}_n^{m-1}\mathbf{U}^{\mathrm{T}}\alpha \\ -\varepsilon\gamma & \varepsilon \end{bmatrix} \end{aligned} \tag{7}$$

$$\varepsilon = \left(\frac{1}{N}\mathbf{U}\mathbf{U}^{\mathrm{T}} - \frac{1}{N}\mathbf{U}\left(\mathbf{X}_n^{m-1}\right)^{\mathrm{T}}\left(\mathbf{R}_n^{m-1}\right)^{-1}\frac{1}{N}\mathbf{X}_n^{m-1}\mathbf{U}^{\mathrm{T}}\right)^{-1} \tag{8}$$

$$\gamma = \frac{1}{N}\mathbf{U}\left(\mathbf{X}_n^{m-1}\right)^{\mathrm{T}}\left(\mathbf{R}_n^{m-1}\right)^{-1}. \tag{9}$$

Fig. 6 describes the schematic of (8), and Fig. 7 describes the schematic of (9). The green part is to highlight the difference from the multiline multiband correlation matrix of the same line (MLMBCM-S).

2) Multiline Multiband Correlation Matrix of the Same Line: The MLMBCM-S is defined as all the pixels in the first n bands of the hyperspectral image in the m line, marked

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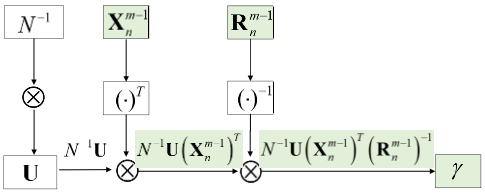


Fig. 7. Schematic of γ in (9).

as $\mathbf{R}_n^m$. Its expression is shown in the following equation:

$$\begin{aligned}\mathbf{R}_n^m &= \frac{1}{N}\mathbf{X}_n^m\left(\mathbf{X}_n^m\right)^{\mathrm{T}} = \frac{1}{N}\begin{bmatrix}\mathbf{X}_{n-1}^m \\ \mathbf{U}\end{bmatrix}\begin{bmatrix}\left(\mathbf{X}_{n-1}^m\right)^{\mathrm{T}} & \mathbf{U}^{\mathrm{T}}\end{bmatrix} \\ &= \frac{1}{N}\begin{bmatrix}\mathbf{X}_{n-1}^m\left(\mathbf{X}_{n-1}^m\right)^{\mathrm{T}} & \mathbf{X}_{n-1}^m\mathbf{U}^{\mathrm{T}} \\ \mathbf{U}\left(\mathbf{X}_{n-1}^m\right)^{\mathrm{T}} & \mathbf{U}\mathbf{U}^{\mathrm{T}}\end{bmatrix} \\ &= \begin{bmatrix}\frac{1}{N}\mathbf{X}_{n-1}^m\left(\mathbf{X}_{n-1}^m\right)^{\mathrm{T}} & \frac{1}{N}\mathbf{X}_{n-1}^m\mathbf{U}^{\mathrm{T}} \\ \frac{1}{N}\mathbf{U}\left(\mathbf{X}_{n-1}^m\right)^{\mathrm{T}} & \frac{1}{N}\mathbf{U}\mathbf{U}^{\mathrm{T}}\end{bmatrix} \\ &= \begin{bmatrix}\mathbf{R}_{n-1}^m & \frac{1}{N}\mathbf{X}_{n-1}^m\mathbf{U}^{\mathrm{T}} \\ \frac{1}{N}\mathbf{U}\left(\mathbf{X}_{n-1}^m\right)^{\mathrm{T}} & \frac{1}{N}\mathbf{U}\mathbf{U}^{\mathrm{T}}\end{bmatrix}. \end{aligned} \tag{10}$$

Anomaly detection can be realized by inverting the correlation matrix $\mathbf{R}_n^m$ of multiple lines and multiple bands, which is consistent with Section II-B1. The block matrix inversion (5) is used for recursive update and solution. The following equation can be obtained from (10):

$$\left(\mathbf{R}_n^m\right)^{-1} = \begin{bmatrix}\mathbf{R}_{n-1}^m & \frac{1}{N}\mathbf{X}_{n-1}^m\mathbf{U}^{\mathrm{T}} \\ \frac{1}{N}\mathbf{U}\left(\mathbf{X}_{n-1}^m\right)^{\mathrm{T}} & \frac{1}{N}\mathbf{U}\mathbf{U}^{\mathrm{T}}\end{bmatrix}^{-1}. \tag{11}$$

Refer to the block matrix of (5) to calculate the inverse equation. Let $\mathbf{E} = \mathbf{R}_{n-1}^m$, $\mathbf{F} = (1/N)\mathbf{X}_{n-1}^m\mathbf{U}^{\mathrm{T}}$, $\mathbf{G} = (1/N)\mathbf{U}(\mathbf{X}_{n-1}^m)^{\mathrm{T}}$, and $\mathbf{H} = (1/N)\mathbf{U}\mathbf{U}^{\mathrm{T}}$, then the recursive update of (11) can be obtained by (12)

$$\begin{aligned}&\left(\mathbf{R}_n^m\right)^{-1} \\ &= \begin{bmatrix}\mathbf{R}_{n-1}^m & \frac{1}{N}\mathbf{X}_{n-1}^m\mathbf{U}^{\mathrm{T}} \\ \frac{1}{N}\mathbf{U}\left(\mathbf{X}_{n-1}^m\right)^{\mathrm{T}} & \frac{1}{N}\mathbf{U}\mathbf{U}^{\mathrm{T}}\end{bmatrix}^{-1} \\ &= \begin{bmatrix}\left(\mathbf{R}_{n-1}^m\right)^{-1}+\frac{1}{N}\left(\mathbf{R}_{n-1}^m\right)^{-1}\mathbf{X}_{n-1}^m\mathbf{U}^{\mathrm{T}}\alpha\beta & -\frac{1}{N}\left(\mathbf{R}_{n-1}^m\right)^{-1}\mathbf{X}_{n-1}^m\mathbf{U}^{\mathrm{T}}\alpha \\ -\alpha\beta & \alpha\end{bmatrix}\end{aligned} \tag{12}$$

$$\alpha = \left(\frac{1}{N}\mathbf{U}\mathbf{U}^{\mathrm{T}} - \frac{1}{N}\mathbf{U}\left(\mathbf{X}_{n-1}^m\right)^{\mathrm{T}}\left(\mathbf{R}_{n-1}^m\right)^{-1}\frac{1}{N}\mathbf{X}_{n-1}^m\mathbf{U}^{\mathrm{T}}\right)^{-1} \tag{13}$$

$$\beta = \frac{1}{N}\mathbf{U}\left(\mathbf{X}_{n-1}^m\right)^{\mathrm{T}}\left(\mathbf{R}_{n-1}^m\right)^{-1}. \tag{14}$$

Fig. 8 describes the schematic of (13), and Fig. 9 describes the schematic of (14). The pink part is to highlight the difference from MLMBCM-D.

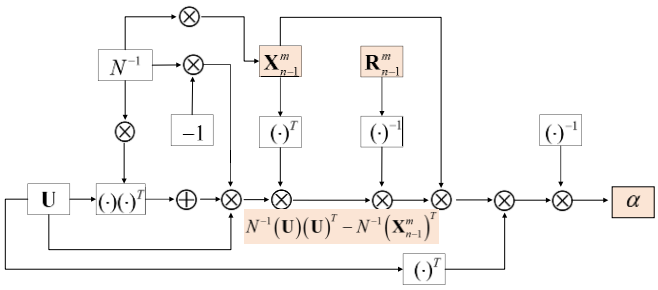


Fig. 8. Schematic of α in (13).

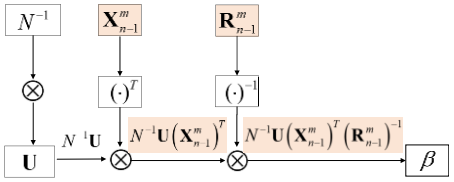


Fig. 9. Schematic of β in (14).

C. RTMLMB-RAD Using MLMBCM

1) RTMLMB-RAD Using MLMBCM-D: The MLMBCM is obtained by using (7)–(9) with different lines. According to R-AD (15), the detection operator of RTMLMB-RAD is shown in (16), as shown at the bottom of the next page

$$\delta^{\text{R-AD}}(\mathbf{r}) = (\mathbf{r})^{\mathrm{T}}(\mathbf{R})^{-1}\mathbf{r} \tag{15}$$

where $\mathbf{r}$ is the sample vector and $\mathbf{R}$ is the correlation matrix. The multiline and multiband detection operator of RTMLMB-RAD with different lines can be defined as (16). Fig. 10 depicts a schematic of (16).

2) RTMLMB-RAD Using MLMBCM-S: Similarly, the correlation matrix of MLMB is obtained by (12)–(14). RTMLMB-RAD using MLMBCM-S can be defined as (18), as shown at the bottom of the next page.

Fig. 11 depicts the schematic implementation of (18). Notably, Fig. 11 shows how the real-time online processing local anomaly detection derived in this article is recursively updated through (18) previously calculated and current line band data.

As can be seen from Figs. 10 and 11, after continuous iteration, the detection process of the current line not only contains different band information of the current line, but also utilizes the local information of the previous line. Therefore, the proposed RTMLMB-RAD is a method of anomaly detection in the complex BKG by using local information. A detailed step-by-step implementation of RTMLMB-RAD is given as follows:

III. Computation and Time Complexity

The computational complexity and the time required to run the local anomaly detection algorithm for hyperspectral images are important indicators to evaluate the performance of the local anomaly detection algorithm. Table I details the calculation complexity of (18) $\delta^{\text{RTMLMB-RAD}}(\mathbf{r}_n^m)$ of the proposed RTMLMB-RAD method, in which the third column specifies the mathematical operation amount of the second column in words, and the fourth column is complexity.

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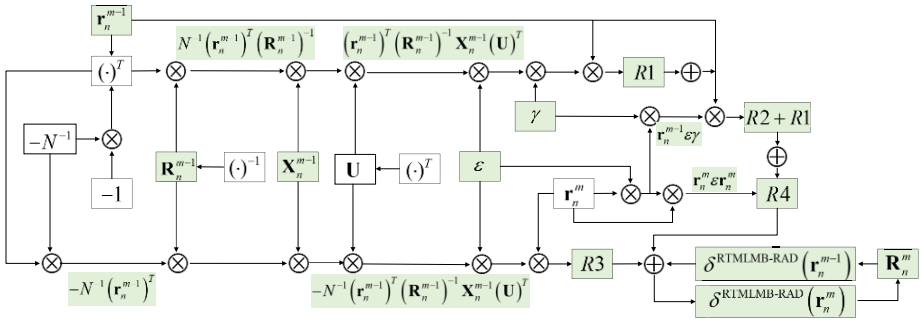


Fig. 10. Schematic of $\delta^{\text{RTMLMB-RAD}}(\mathbf{r}_n^m)$ between different lines in (16).

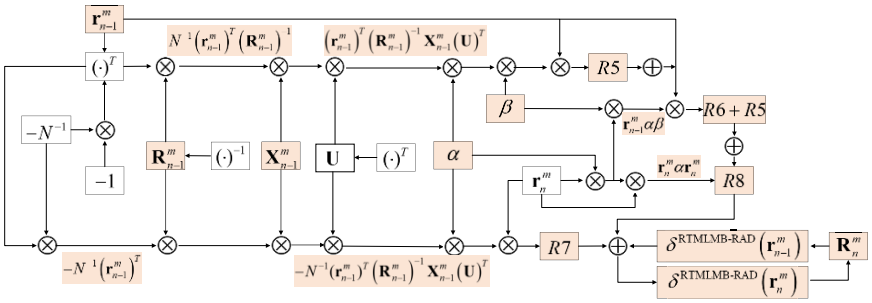


Fig. 11. Schematic of $\delta^{\text{RTMLMB-RAD}}(\mathbf{r}_n^m)$ of the same line in (18).

Table II shows the comparison of the computational complexity of the R-AD and the proposed RTMLMB-RAD method for processing a single signature. It can be seen that the complexity of matrix inversion of the R-AD method is $O(n^3)$, while the real-time RTMLMB-RAD method proposed in this article uses the recursive updating formula of the block matrix

$$
\begin{aligned}
&\delta^{\text{RTMLMB-RAD}}\left(\mathbf{r}_n^m\right)\\
&=\left(\mathbf{r}_n^m\right)^{\mathrm{T}}(\mathbf{R})^{-1}\mathbf{r}_n^m=\left(\left(\mathbf{r}_n^{m-1}\right)^{\mathrm{T}},\mathbf{r}_n^m\right)\begin{bmatrix}\left(\mathbf{R}_n^{m-1}\right)^{-1}+\dfrac{1}{N}\left(\mathbf{R}_n^{m-1}\right)^{-1}\mathbf{X}_n^{m-1}\mathbf{U}^{\mathrm{T}}\alpha\beta & -\dfrac{1}{N}\left(\mathbf{R}_n^{m-1}\right)^{-1}\mathbf{X}_n^{m-1}\mathbf{U}^{\mathrm{T}}\alpha\\ -\alpha\beta & \alpha\end{bmatrix}\begin{pmatrix}\mathbf{r}_n^{m-1}\\ \mathbf{r}_n^m\end{pmatrix}\\
&=\left(\mathbf{r}_n^{m-1}\right)^{\mathrm{T}}\left(\left(\mathbf{R}_n^{m-1}\right)^{-1}+\frac{1}{N}\left(\mathbf{R}_n^{m-1}\right)^{-1}\mathbf{X}_n^{m-1}\mathbf{U}^{\mathrm{T}}\alpha\beta\right)\mathbf{r}_n^{m-1}-\mathbf{r}_n^m\alpha\beta\mathbf{r}_n^{m-1}+\mathbf{r}_n^m\alpha\mathbf{r}_n^m-\frac{1}{N}\left(\mathbf{r}_n^{m-1}\right)^{\mathrm{T}}\left(\mathbf{R}_n^{m-1}\right)^{-1}\mathbf{X}_n^{m-1}\mathbf{U}^{\mathrm{T}}\alpha\mathbf{r}_n^m\\
&=\left(\mathbf{r}_n^{m-1}\right)^{\mathrm{T}}\left(\mathbf{R}_n^{m-1}\right)^{-1}\mathbf{r}_n^{m-1}+\frac{1}{N}\left(\mathbf{r}_n^{m-1}\right)^{\mathrm{T}}\left(\mathbf{R}_n^{m-1}\right)^{-1}\mathbf{X}_n^{m-1}\mathbf{U}^{\mathrm{T}}\alpha\beta\mathbf{r}_n^{m-1}-\mathbf{r}_n^m\alpha\beta\mathbf{r}_n^{m-1}+\mathbf{r}_n^m\alpha\mathbf{r}_n^m-\frac{1}{N}\left(\mathbf{r}_n^{m-1}\right)^{\mathrm{T}}\left(\mathbf{R}_n^{m-1}\right)^{-1}\mathbf{X}_n^{m-1}\mathbf{U}^{\mathrm{T}}\alpha\mathbf{r}_n^m\\
&=\delta^{\text{RTMLMB-RAD}}\left(\mathbf{r}_n^{m-1}\right)+\frac{1}{N}\left(\mathbf{r}_n^{m-1}\right)^{\mathrm{T}}\left(\mathbf{R}_n^{m-1}\right)^{-1}\mathbf{X}_n^{m-1}\mathbf{U}^{\mathrm{T}}\alpha\beta\mathbf{r}_n^{m-1}-\mathbf{r}_n^m\alpha\beta\mathbf{r}_n^{m-1}+\mathbf{r}_n^m\alpha\mathbf{r}_n^m-\frac{1}{N}\left(\mathbf{r}_n^{m-1}\right)^{\mathrm{T}}\left(\mathbf{R}_n^{m-1}\right)^{-1}\mathbf{X}_n^{m-1}\mathbf{U}^{\mathrm{T}}\alpha\mathbf{r}_n^m
\end{aligned}
\tag{16}
$$

$$
\begin{aligned}
&\delta^{\text{RTMLMB-RAD}}\left(\mathbf{r}_n^m\right)\\
&=\delta^{\text{RTMLMB-RAD}}\left(\mathbf{r}_n^{m-1}\right)+R1+R2+R3+R4
\end{aligned}
\tag{17}
$$

$$
\begin{aligned}
&\delta^{\text{RTMLMB-RAD}}\left(\mathbf{r}_n^m\right)\\
&=\left(\mathbf{r}_n^m\right)^{\mathrm{T}}(\mathbf{R})^{-1}\mathbf{r}_n^m=\left(\left(\mathbf{r}_{n-1}^m\right)^{\mathrm{T}},\mathbf{r}_n^m\right)\begin{bmatrix}\left(\mathbf{R}_{n-1}^m\right)^{-1}+\dfrac{1}{N}\left(\mathbf{R}_{n-1}^m\right)^{-1}\mathbf{X}_{n-1}^m\mathbf{U}^{\mathrm{T}}\alpha\beta & -\dfrac{1}{N}\left(\mathbf{R}_{n-1}^m\right)^{-1}\mathbf{X}_{n-1}^m\mathbf{U}^{\mathrm{T}}\alpha\\ -\alpha\beta & \alpha\end{bmatrix}\begin{pmatrix}\mathbf{r}_{n-1}^m\\ \mathbf{r}_n^m\end{pmatrix}\\
&=\left(\mathbf{r}_{n-1}^m\right)^{\mathrm{T}}\left(\left(\mathbf{R}_{n-1}^m\right)^{-1}+\frac{1}{N}\left(\mathbf{R}_{n-1}^m\right)^{-1}\mathbf{X}_{n-1}^m\mathbf{U}^{\mathrm{T}}\alpha\beta\right)\mathbf{r}_{n-1}^m-\mathbf{r}_n^m\alpha\beta\mathbf{r}_{n-1}^m+\mathbf{r}_n^m\alpha\mathbf{r}_n^m-\frac{1}{N}\left(\mathbf{r}_{n-1}^m\right)^{\mathrm{T}}\left(\mathbf{R}_{n-1}^m\right)^{-1}\mathbf{X}_{n-1}^m\mathbf{U}^{\mathrm{T}}\alpha\mathbf{r}_n^m\\
&=\left(\mathbf{r}_{n-1}^m\right)^{\mathrm{T}}\left(\mathbf{R}_{n-1}^m\right)^{-1}\mathbf{r}_{n-1}^m+\frac{1}{N}\left(\mathbf{r}_{n-1}^m\right)^{\mathrm{T}}\left(\mathbf{R}_{n-1}^m\right)^{-1}\mathbf{X}_{n-1}^m\mathbf{U}^{\mathrm{T}}\alpha\beta\mathbf{r}_{n-1}^m-\mathbf{r}_n^m\alpha\beta\mathbf{r}_{n-1}^m+\mathbf{r}_n^m\alpha\mathbf{r}_n^m-\frac{1}{N}\left(\mathbf{r}_{n-1}^m\right)^{\mathrm{T}}\left(\mathbf{R}_{n-1}^m\right)^{-1}\mathbf{X}_{n-1}^m\mathbf{U}^{\mathrm{T}}\alpha\mathbf{r}_n^m\\
&=\delta^{\text{RTMLMB-RAD}}\left(\mathbf{r}_{n-1}^m\right)+\frac{1}{N}\left(\mathbf{r}_{n-1}^m\right)^{\mathrm{T}}\left(\mathbf{R}_{n-1}^m\right)^{-1}\mathbf{X}_{n-1}^m\mathbf{U}^{\mathrm{T}}\alpha\beta\mathbf{r}_{n-1}^m-\mathbf{r}_n^m\alpha\beta\mathbf{r}_{n-1}^m+\mathbf{r}_n^m\alpha\mathbf{r}_n^m-\frac{1}{N}\left(\mathbf{r}_{n-1}^m\right)^{\mathrm{T}}\left(\mathbf{R}_{n-1}^m\right)^{-1}\mathbf{X}_{n-1}^m\mathbf{U}^{\mathrm{T}}\alpha\mathbf{r}_n^m
\end{aligned}
\tag{18}
$$

$$
\begin{aligned}
&\delta^{\text{RTMLMB-RAD}}\left(\mathbf{r}_n^m\right)\\
&=\delta^{\text{RTMLMB-RAD}}\left(\mathbf{r}_{n-1}^m\right)+R5+R6+R7+R8
\end{aligned}
\tag{19}
$$

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TABLE I
COMPUTATIONAL COMPLEXITY OF (11) TO CALCULATE

Initial conditions	$\mathbf{U}$	$\mathbf{U}\mathbf{U}^T$	$\mathbf{U}$ (n multiplications)	N/A
Innovations information: α by (13)		$\alpha=\left(\frac{1}{N}\mathbf{U}\mathbf{U}^T-\frac{1}{N}\mathbf{U}(\mathbf{X}_{n-1}^m)^T\left(\mathbf{R}_{n-1}^m\right)^{-1}\frac{1}{N}\mathbf{X}_{n-1}^m\mathbf{U}^T\right)^{-1}$	One inner product of an n-dimensional vector $\mathbf{U}\mathbf{U}^T$ (n multiplications) and one inner product	$O(n)$
Innovations information: β by (14)		$\beta=\frac{1}{N}\mathbf{U}(\mathbf{X}_{n-1}^m)^T\left(\mathbf{R}_{n-1}^m\right)^{-1}$	Inner products of two n-dimensional vectors and two inner products	$O(n)$
Update $R5+R6+R7+R8$ by (18)		$R5+R6+R7+R8=\frac{1}{N}\left(\mathbf{r}_{n-1}^m\right)^T\left(\mathbf{R}_{n-1}^m\right)^{-1}\mathbf{X}_{n-1}^m\mathbf{U}^T\alpha\beta\mathbf{r}_{n-1}^m-\mathbf{r}_n^m\alpha\beta\mathbf{r}_{n-1}^m$ $-\frac{1}{N}\left(\mathbf{r}_{n-1}^m\right)^T\left(\mathbf{R}_{n-1}^m\right)^{-1}\mathbf{X}_{n-1}^m\mathbf{U}^T\alpha\mathbf{r}_n^m+\mathbf{r}_n^m\alpha\mathbf{r}_n^m$	Seven inner products of n-dimensional vectors + four inner products of n-dimensional vectors plus 1 subtraction and 1multiplication	$O(n)$
Update $\delta^{\text{RTMRB-RAD}}(\mathbf{r}_n^m)$ by (13), (14), and (18)		$\delta^{\text{RTMLMB-RAD}}(\mathbf{r}_n^m)=\delta^{\text{RTMLMB-RAD}}(\mathbf{r}_{n-1}^m)+R5+R6+R7+R8$	Inner products of two n-dimensional vectors + Seven inner products of n-dimensional vectors + four inner products of n-dimensional vectors plus 1 subtraction and 1 multiplication	$O(n)$

Algorithm 1 Algorithm RTMLMB-RAD

Input: Hyperspectral image.

Step 1. Initialization: Initialize the number of bands n determined by the number of filters, initialize the number of rows j extracted based on the band, and initialize the number of intervals k between different bands.

Step 2. Transformation: According to the above n, j and k values, the data is converted into data requiring real-time detection processing.

Step 3. Calculate: According to n value, establish the state equation of spectral vector multi-line multi-band correlation matrix R. Then compute the anomalous degree of each pixel.

If $n=1$, calculate R by (4).

Update the inverse of the correlation matrix R, calculate R^{-1} by (7).

Compute the anomalous degree of each pixel: calculate $\delta^{\text{RTMLMB-RAD}}(\mathbf{r}_n^m)$ by (16).

Elseif $n>1$, calculate R by (10).

Update the inverse of the correlation matrix R, calculate R^{-1} by (12).

Compute the anomalous degree of each pixel: calculate $\delta^{\text{RTMLMB-RAD}}(\mathbf{r}_n^m)$ by (18).

End

Update n: $n\leftarrow n+1$

Output: Anomaly map.

TABLE II
COMPARISON OF COMPUTATION COMPLEXITY OF MATRIX INVERSIONS BY PROCESSING A SINGLE SIGNATURE

	R-AD specified by (15)	RTMLMB-RAD specified by (18)
Input	$\mathbf{U}_1,\mathbf{U}_2,...,\mathbf{U}_n$	$\mathbf{U}$
The complexity of method	$O(n^3)$	$O(n)$

inversion formula, and the addition and subtraction operations with low matrix complexity avoid a large number of matrix inversion.

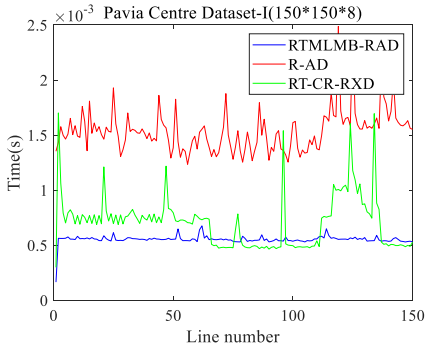


Fig. 12. Comparison of calculation time required by RTMLMB-RAD and other algorithms in Pavia Centre Dataset-I data.

TABLE III
COMPARATIVE PERFORMANCE OF AVERAGE COMPUTING TIMES REQUIRED BY RTMLMB-RAD, R-AD, AND RT-CR-RXD

Method	Pavia Centre Dataset-I
RTMLMB-RAD (seconds)	**0.0006**
R-AD (seconds)	0.0015
RT-CR-RXD (seconds)	0.0007

To verify the timeliness of the proposed RTMLMB-RAD in this article, a real hyperspectral image data experiment was conducted with the R-AD and real-time causal R-RXD (RT-CR-RXD) [31] algorithms, as shown in Fig. 12. Among them, the reason why the R-AD algorithm is selected for comparison is that R-AD algorithm is a typical anomaly detection algorithm. The RT-CR-RXD algorithm is chosen because it is a real-time anomaly detection algorithm. In order to verify the timeliness, the uniform band selection (UBS) method is used to process the real hyperspectral image Pavia Centre Dataset-I, and the data size is changed to 150 × 150 × 8. The detailed introduction of Pavia Centre Dataset-I data is shown in the experiment. Fig. 12 shows the single-line calculation time required by the three methods when the number of detected data lines increases from 1 to 150, where the x-axis represents the number of data lines.

As can be seen from Table III, the proposed RTMLMB-RAD method has less computation time than the R-AD algorithm and RT-CR-RXD algorithm. On the

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whole, with the increase in the number of lines, the calculation time required by RTMLMB-RAD tends to be stable, and it only requires almost constant calculation time to process each signal source, which is less than the R-AD algorithm and RT-CR-RXD algorithm, indicating that the proposed algorithm has better timeliness.

IV. Experimental Results

In this section, 16 real hyperspectral data captured in different scenarios are used to evaluate the effectiveness and superiority of the proposed RTMLMB-RAD. The 16 real hyperspectral data include six widely used anomaly detection hyperspectral data and ten large scene industrial tobacco scenario hyperspectral data. At the same time, the proposed algorithm is compared with six anomaly detection methods. The first is RXD [15], which is recognized as the benchmark of anomaly detection methods. The second is the R-AD method for anomaly detection based on a correlation matrix with causality characteristics [16], and the response degree of the proposed algorithm to anomaly targets can be seen by comparison. The third FrFE-RX method for anomaly detection is to enhance the distinction between anomaly and BKG by removing noise [18]. The proposed algorithm can be compared with the FrFE-RX method to reflect the ability to distinguish targets and BKGs. The fourth is the anomaly detection local summation unsupervised nearest regularized subspace with an outlier removal anomaly detector (LSUNRSORAD) method which makes full use of the local BKG of each local window of the hyperspectral image pixel [42]. The local anomaly detection ability can be reflected by comparing the proposed algorithm with it. According to the three parameter values adopted by LSUNRSORAD algorithm, this paper uniformly set $\text{win}_{\text{out}} = 5$, $\text{win}_{\text{in}} = 3$, $\lambda = 0.01$ in the comparison experiment. The fifth is RT-CR-RXD [31], which is a typical real-time anomaly detection method based on a causality sampling correlation matrix. The sixth method is progressive band subset fusion real time (PBSF-RT) [43], which is a real-time anomaly detection method based on progressive band subset fusion. It obtains data by band-by-band. The ability of real-time detection can be seen by comparing the proposed algorithm with it. In addition, to avoid detection errors, the band selected by the comparison method is consistent with that of the proposed method. It is worth noting that when verifying the applicability of the proposed method on the six datasets, the data are rearranged according to the way that the multiarray filter spectral imaging camera collects. However, none of the six comparison methods can directly process the data obtained by the multiarray filter spectral imaging camera, and the detection should be performed after the complete data are obtained. Therefore, to a certain extent, the proposed algorithm has reflected the application advantages of a multiarray filter spectral imaging system.

In the experiment, the detection performance was evaluated in qualitative and quantitative ways. In particular, the detection diagram of each detection method is given, which qualitatively evaluates the detection performance. The 3-D receiver operating characteristic (3-D ROC) curve [44], [45], 2-D receiver operating characteristic (2-D ROC) curve, and area under the ROC curve (AUC) were used to quantitatively evaluate the detection performance [46], [47]. Among them, the 3-D ROC curve generates (P_{D}, P_{F}, τ) function to evaluate the detection performance by introducing specific threshold parameter τ as the third independent variable. Three types of 2-D ROC curves include the 2-D ROC curve of (P_{D}, P_{F}), the 2-D ROC curve of (P_{D}, τ), and the 2-D ROC curve of (P_{F}, τ). The AUC value is the AUC surrounded by the axes. The higher the AUC of (P_{D}, P_{F}) and (P_{D}, τ) values, the better the target detectability. The AUC of (P_{F}, τ) is used to evaluate BKG inhibition. The smaller the AUC of (P_{F}, τ), the better the inhibition of BKG.

All experimental algorithms were implemented using MATLAB R2020a on a PC equipped with Intel Core i5-8265 CPU at 1.8 GHz and 8-GB RAM.

A. Dataset Description

1) Pavia Centre Dataset-I: The first dataset was obtained using the Reflective Optics System Imaging Spectrometer (ROSIS) sensor in the city of Pavia, Italy. It has a spatial resolution of 1.3 m. The spectral information of eight bands was retained for subsequent experiments by the UBS method, and the selected representative band subset was [30, 40, 50, 60, 70, 80, 90, 100]. As the research focus of this article is anomaly detection, band selection is not introduced here, and the following data description will not be repeated. The whole image contains 1096 × 1096 pixels. A subscene of 150 × 150 pixels is selected for anomaly detection. Its BKG mainly includes bridges, water, bare soil, and buildings, and vehicles on the bridge are regarded as anomalies to be detected. The false-color image, ground truth, and spectral signature of the target are shown in Fig. 13(a).

2) Pavia Centre Dataset-II: Compared with Pavia Centre Dataset-I, the second dataset selected a 150 × 150 pixel subscene for anomaly detection and expressed as Pavia Centre Dataset-II. The BKG of the subscene mainly includes the water and the beach, and the airplanes are the anomalies to be detected. The false-color image, ground truth, and spectral signature of the target are shown in Fig. 13(b).

3) Gulfport Dataset: The third dataset was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) in the Gulfport Airport area of MS, USA. It contains 191 bands in the wavelength range of 550–1850 nm, with a spatial resolution of 3.4 m. The subscene with 100 × 100 pixels is selected for anomaly detection. Its BKG is mainly the airport runway, highway, and some vegetation. Three aircraft with different sizes are the anomalies to be detected. The false-color image, ground truth, and spectral signature of the target are shown in Fig. 13(c).

4) Los Angeles Dataset-I: The fourth dataset was collected by AVIRIS in Los Angeles, CA, USA. It has 205 bands distributed in the wavelength range of 430–860, with a spatial resolution of 7.1 m. Select a subscene of 100 × 100 pixels for anomaly detection, and the aircraft is regarded as the anomaly to be detected. The false-color image, ground truth, and spectral signature of the target are shown in Fig. 13(d).

5) Los Angeles Dataset II: Compared with Los Angeles Dataset I, the fifth dataset was collected at another location in Los Angeles. Select 100 × 100 pixels subscene is used

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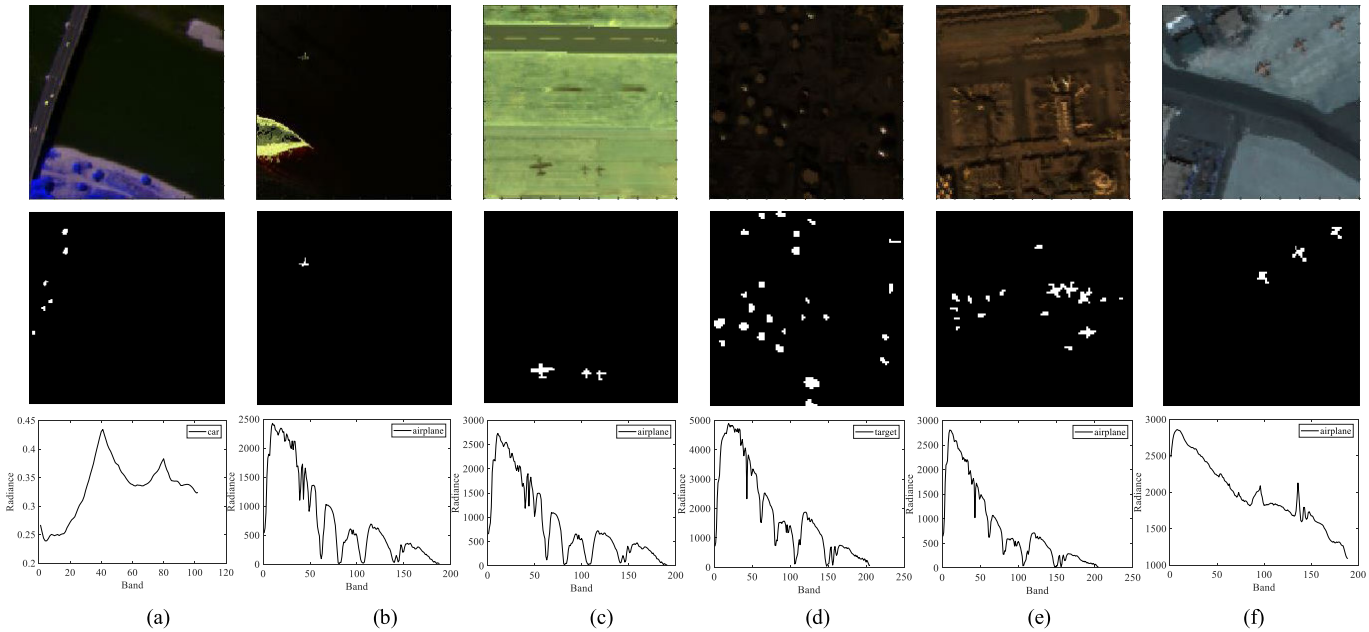


Fig. 13. Pseudocolor images, ground-truth maps, and spectral signature of six experimental datasets: (a) Pavia Centre-I; (b) Pavia Centre-II; (c) Gulfport; (d) Los Angeles-I; (e) Los Angeles-II; and (f) San Diego.

for anomaly detection, and the airplane is considered as the anomaly to be detected. The false-color image, ground truth, and spectral signature of the target are shown in Fig. 13(e).

6) San Diego Dataset: The sixth dataset was captured by AVIRIS in the airport area of San Diego, CA. It consists of 224 bands, with wavelengths ranging from 370 to 2510 nm, and a spatial resolution of 3.5 m. The whole image contains 400 × 400 pixels. A subscene of 100 × 100 pixels is selected from the top left corner of the image for anomaly detection, and its BKG is mainly the hangar, apron, and bare soil. Three planes at the airport are being treated as anomalies for testing. The false-color image, ground truth, and spectral signature of the target are shown in Fig. 13(f).

B. Detection Performance

In this section, the performance of the proposed RTMLMB-RAD is evaluated by comparing it with six other state-of-the-art anomaly detection methods. First of all, we made a qualitative study of the detection performance through the detection diagram. Fig. 14 shows the detection process of the proposed RTMLMB-RAD on six real hyperspectral datasets, including detection maps and grayscale maps. It can be observed that the proposed RTMLMB-RAD can accurately detect the positions and shapes of different size anomalies in different scenes such as aircraft and vehicles. This is because RTMLMB-RAD uses local results from previous and current data. The local method can retain the target better, but at the same time, the suppression effect of the BKG is slightly worse than that of the global method. It is worth noting that when there are large differences in local areas, the proposed RTMLMB-RAD can better play its inherent advantages and achieve a good response degree to the anomaly target.

The detection results of the proposed method on the six datasets are shown in Fig. 15. Compared with other methods, the proposed RTMLMB-RAD is inclined to reserve the anomalies in the detection graph. As can be seen from Fig. 15, RXD, R-AD, FrFE-RX, LSUNRSORAD, and PBSF-RT achieve better BKG suppression for Pavia Centre Dataset-I and Pavia Centre Dataset II datasets. However, the detection of anomalies is not complete, but the proposed method can achieve the corresponding anomalies more completely by comparison. In Fig. 15, for the Gulfport dataset, the strong interferences with light reflectance at the top of the image are suppressed better by the proposed RTMLMB-RAD, which can accurately detect the position and shape of the three aircraft as well. However, RXD, R-AD, LSUNRSORAD, and PBSF-RT failed to detect the aircraft, and FrFE-RX and RT-CR-RXD failed to detect the other two small aircrafts, while some undesired regions were also incorrectly detected as anomalies. From Fig. 15, it can be seen that for Los Angeles-I and Los Angeles-II datasets, the proposed RTMLMB-RAD accurately detects the location and shape of all anomalies. However, RXD, R-AD, FrFE-RX, LSUNRSORAD, RT-CR-RXD, and PBSF-RT anomaly targets are almost completely ignored and buried in the BKG. In addition, as shown in Fig. 15, for the San Diego dataset, RTMLMB-RAD can reserve the position and shape of anomalies, which were almost all missed by RXD, R-AD, FrFE-RX, RT-CR-RXD, and PBSF-RT, and at the same time, some false targets are highlighted. In general, from all detection graphs, the proposed RTMLMB-RAD provides the most differentiated detection results closest to ground truth on all datasets and is superior to the other six comparison methods, as the local similarity of BKG is effectively considered.

In addition, to conduct quantitative research on detection performance, the ROC curves of each method on the Pavia Centre-I dataset are plotted in Fig. 16. According to the 3-D ROC curve and corresponding 2-D ROC curve in Fig. 16, under the same conditions, the proposed RTMLMB-RAD

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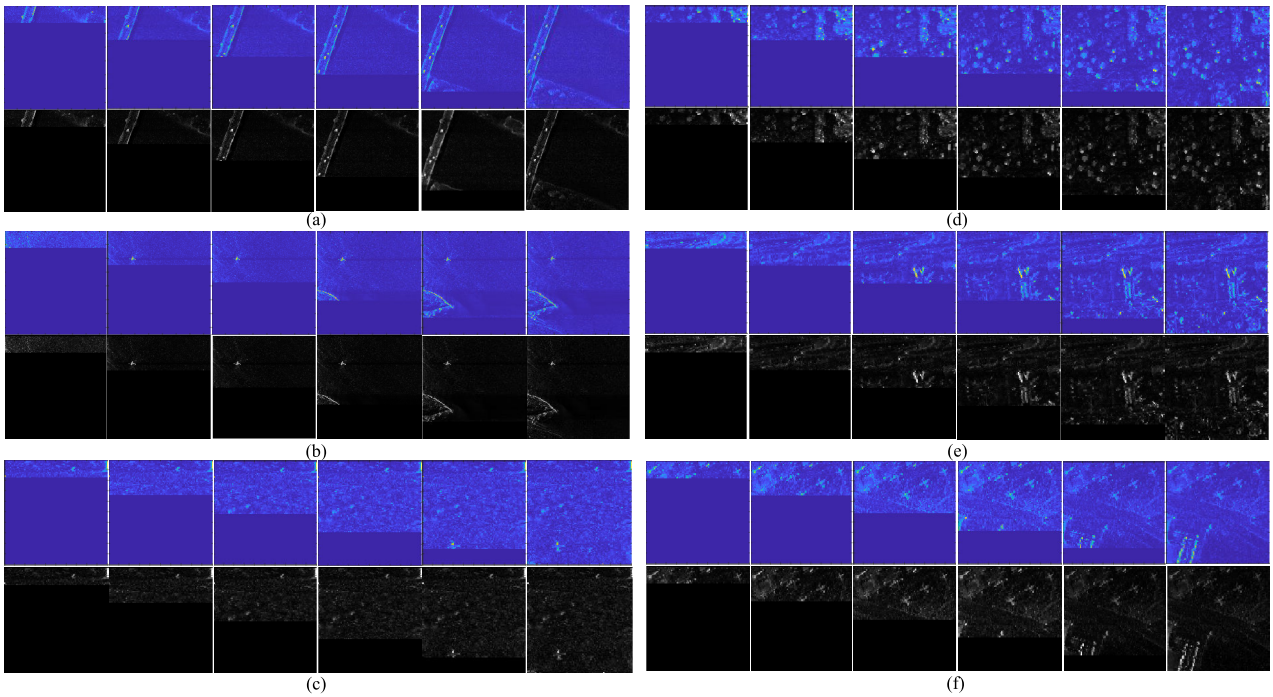


Fig. 14. Detection process of RTMLMB-RAD on six experimental datasets: (a) Pavia Centre-I; (b) Pavia Centre-II; (c) Gulfport; (d) Los Angeles-I; (e) Los Angeles-II; and (f) San Diego.

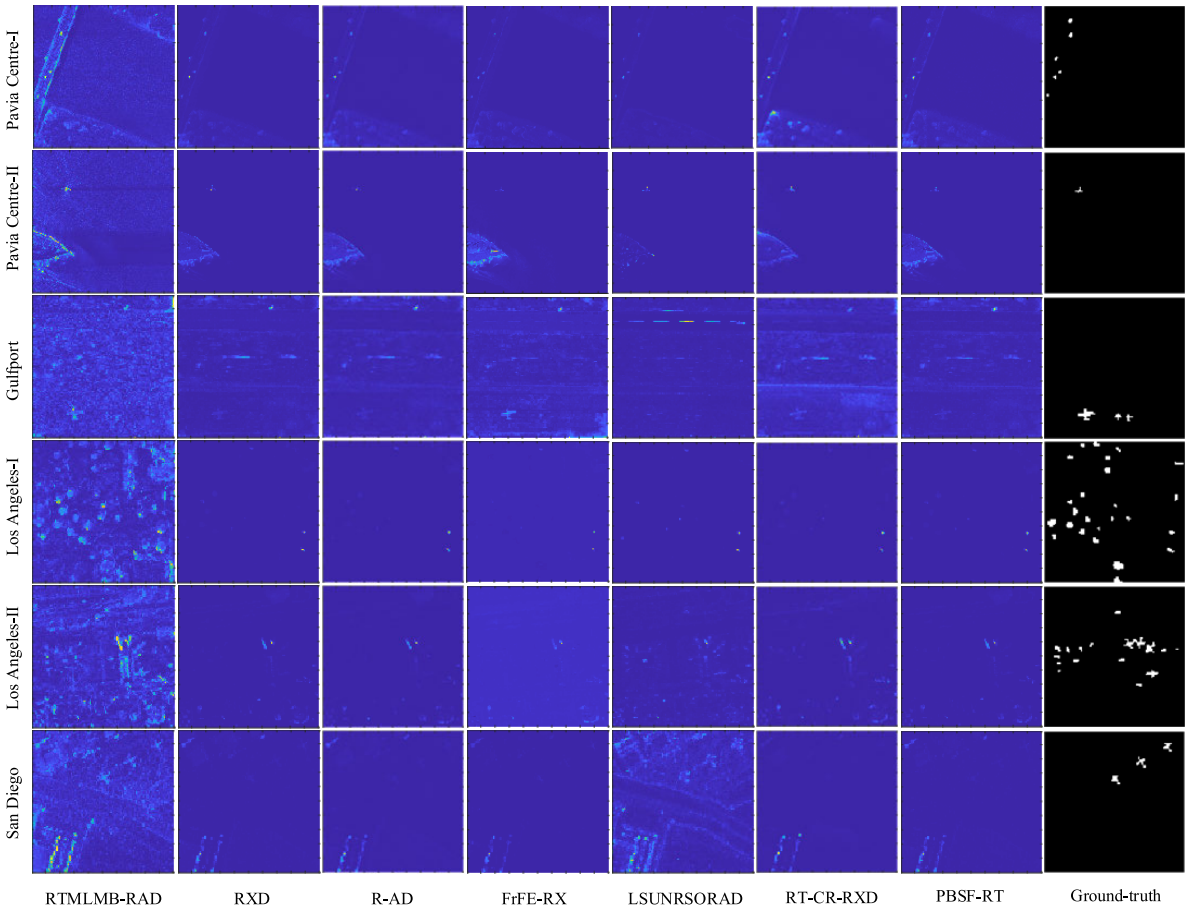


Fig. 15. Detection maps of different methods on six experimental datasets.

always has the highest detection rate. Specifically, for the Pavia Centre-I dataset, the 2-D ROC curve (P_D, P_F) of RTMLMB-RAD bends more to the top left, while the 2-D ROC curve (P_D, τ) bends more to the top right, indicating that the abnormal targets detected have stronger retention ability, and the proposed algorithm has better detection performance.

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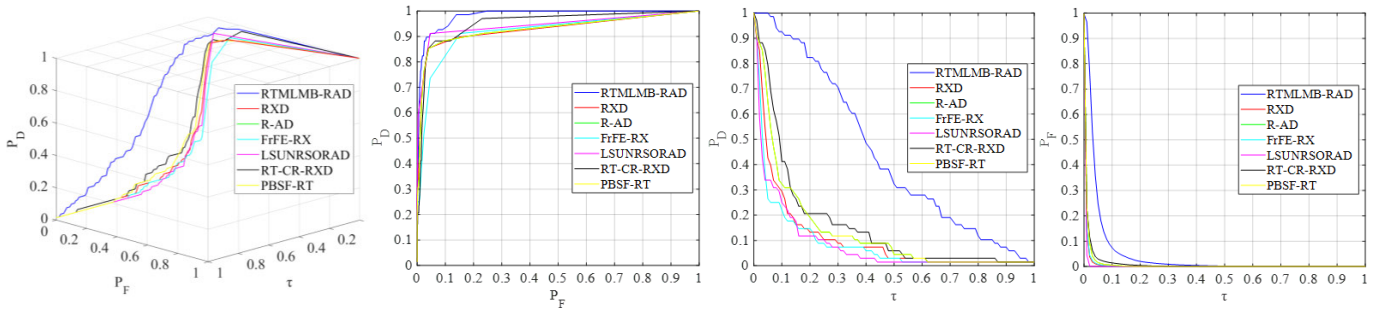


Fig. 16. ROC curves of different methods on Pavia Centre-I: (a) 3-D ROC curves (P_D, P_F, τ); (b) corresponding 2-D ROC curves (P_D, P_F); (c) corresponding 2-D ROC curves (P_D, τ); and (d) corresponding 2-D ROC curves (P_F, τ).

TABLE IV
AUC VALUES OF DIFFERENT METHODS ON PAVIA CENTRE-I

Method	AUC (P_D, P_F)	AUC (P_D, τ)	AUC (P_F, τ)
RTMLMB-RAD	**0.9798**	**0.4463**	0.0458
RXD	0.9259	0.1068	**0.0074**
R-AD	0.9271	0.1349	**0.0097**
FrFE-RX	0.9168	0.0886	0.0076
LSUNRSORAD	0.9427	0.0849	0.0055
RT-CR-RXD	**0.9499**	**0.1596**	0.0121
PBSF-RT	0.9271	0.1349	**0.0097**

TABLE V
AUC VALUES OF DIFFERENT METHODS ON PAVIA CENTRE-II

Method	AUC (P_D, P_F)	AUC (P_D, τ)	AUC (P_F, τ)
RTMLMB-RAD	**0.9536**	**0.4050**	0.0403
RXD	0.8987	0.1005	**0.0071**
R-AD	0.8976	0.1211	0.0082
FrFE-RX	0.9117	0.0913	0.0112
LSUNRSORAD	0.9427	0.1205	**0.0055**
RT-CR-RXD	**0.9656**	**0.2529**	0.0073
PBSF-RT	0.8976	0.1211	0.0082

TABLE VI
AUC VALUES OF DIFFERENT METHODS ON GULFPORT

Method	AUC (P_D, P_F)	AUC (P_D, τ)	AUC (P_F, τ)
RTMLMB-RAD	**0.9225**	**0.2360**	0.0726
RXD	0.8034	0.0327	**0.0172**
R-AD	0.7803	0.0335	0.0184
FrFE-RX	**0.8572**	**0.3215**	0.2186
LSUNRSORAD	0.8561	0.0293	**0.0129**
RT-CR-RXD	0.8156	0.0682	0.0340
PBSF-RT	0.7803	0.0335	0.0184

TABLE VII
AUC VALUES OF DIFFERENT METHODS ON LOS ANGELES-I

Method	AUC (P_D, P_F)	AUC (P_D, τ)	AUC (P_F, τ)
RTMLMB-RAD	**0.9046**	**0.2668**	0.0590
RXD	0.5752	0.0184	**0.0049**
R-AD	0.5803	0.0184	**0.0049**
FrFE-RX	0.3300	0.0262	**0.0223**
LSUNRSORAD	0.5558	0.0182	**0.0049**
RT-CR-RXD	**0.6160**	**0.0224**	**0.0049**
PBSF-RT	0.5803	0.0184	**0.0049**

However, the 2-D ROC curve (P_F, τ) of the RXD method bends more to the bottom left, indicating that its BKG inhibition ability is stronger. In addition, the AUC values of each method on the Pavia Centre-I dataset are shown in Table IV, where red represents the best and green is the second best. It can be observed that the AUC (P_D, P_F) and AUC (P_D, τ) values of RTMLMB-RAD are higher than those of the comparison method, which are 0.9798 and 0.4463, respectively. Although the AUC (P_F, τ) value of RTMLMB-RAD is not the best, it is acceptable. These results show that the proposed RTMLMB-RAD has a higher detection rate.

The ROC curve of each method on the Pavia Centre-II dataset is shown in Fig. 17. From the 3-D ROC curve and corresponding 2-D ROC curve in Fig. 17, it is obvious that the proposed RTMLMB-RAD has good comprehensive performance. As shown in Table V, the AUC (P_D, τ) values of the proposed RTMLMB-RAD are significantly better than those of other comparison methods, and the AUC (P_D, P_F) values of RTMLMB-RAD are second only to RT-CR-RXD. In addition, the AUC (P_F, τ) values of the proposed RTMLMB-RAD algorithm are close to the optimal values.

The ROC curve of each method on the Gulfport dataset is shown in Fig. 18. According to the 3-D ROC curve and corresponding 2-D ROC curve in Fig. 18, the comprehensive performance of the proposed RTMLMB-RAD is very good. As shown in Table VI, the AUC (P_D, P_F) value of the proposed RTMLMB-RAD is better than those of the other six comparison methods, and the AUC (P_D, τ) values of RTMLMB-RAD are second only to FrFE-RX. In addition, the AUC (P_F, τ) value of the proposed RTMLMB-RAD is close to the optimal value, and better than FrFE-RX.

The ROC curve of each method on the Los Angeles-I dataset is shown in Fig. 19. According to the 3-D ROC curve and corresponding 2-D ROC curve in Fig. 19, the comprehensive performance of the proposed RTMLMB-RAD is very good. As shown in Table VII, the AUC (P_D, P_F) and AUC (P_D, τ) values of the proposed RTMLMB-RAD are significantly better than those of other comparison methods.

The ROC curves of each method on the Los Angeles-II dataset are shown in Fig. 20. According to the 3-D ROC curve in Fig. 20 and the corresponding 2-D ROC curve, the proposed RTMLMB-RAD has a good comprehensive performance. As shown in Table VIII, the AUC (P_D, P_F) value of the proposed RTMLMB-RAD is better than other comparison methods, only to be defeated by the LSUNRSORAD method. This is because abnormal targets in the Los Angeles-II dataset are relatively concentrated in the same area and there are

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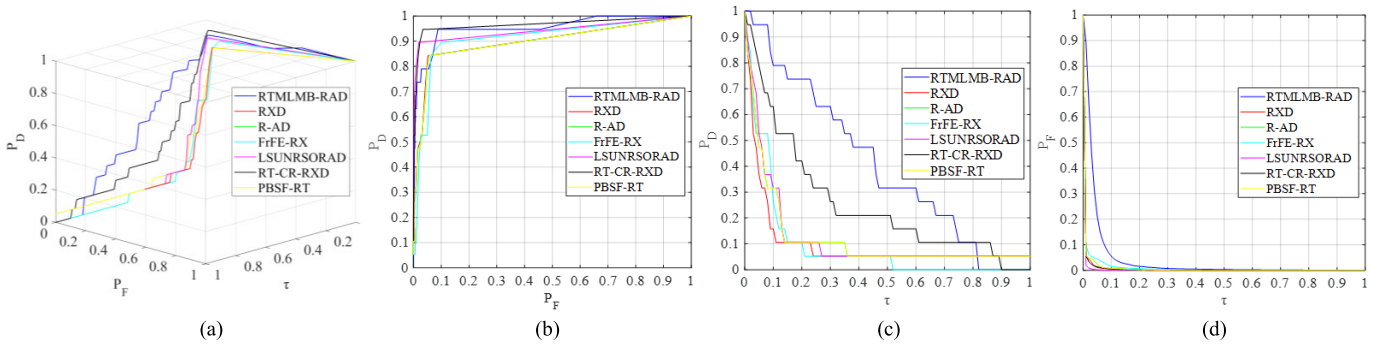


Fig. 17. ROC curves of different methods on Pavia Centre-II: (a) 3-D ROC curves (P_D, P_F, τ); (b) corresponding 2-D ROC curves (P_D, P_F); (c) corresponding 2-D ROC curves (P_D, τ); and (d) corresponding 2-D ROC curves (P_F, τ).

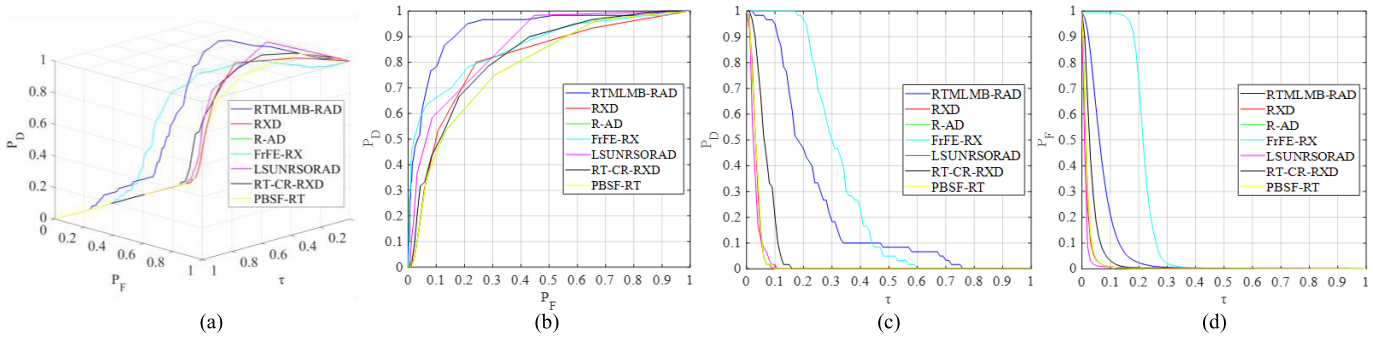


Fig. 18. ROC curves of different methods on Gulfport: (a) 3-D ROC curves (P_D, P_F, τ); (b) corresponding 2-D ROC curves (P_D, P_F); (c) corresponding 2-D ROC curves (P_D, τ); and (d) corresponding 2-D ROC curves (P_F, τ).

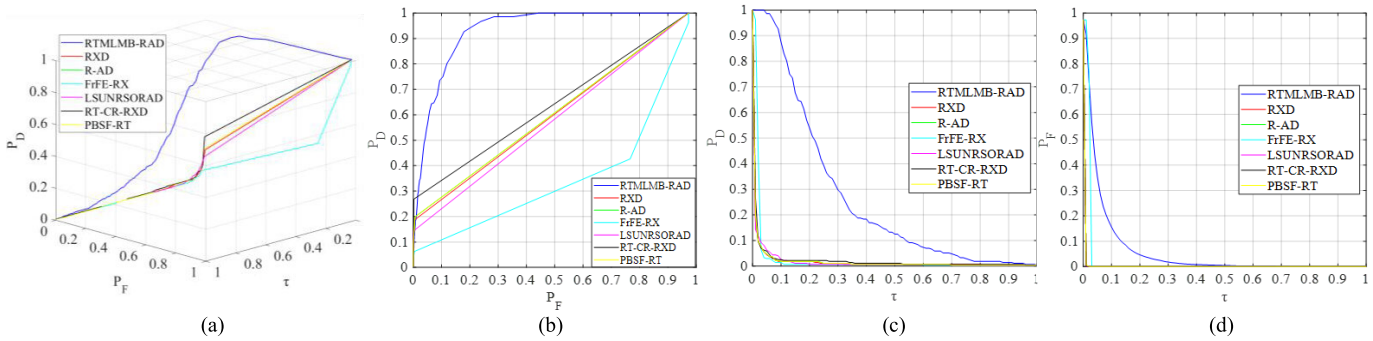


Fig. 19. ROC curves of different methods on Los Angeles-I: (a) 3-D ROC curves (P_D, P_F, τ); (b) corresponding 2-D ROC curves (P_D, P_F); (c) corresponding 2-D ROC curves (P_D, τ); and (d) corresponding 2-D ROC curves (P_F, τ).

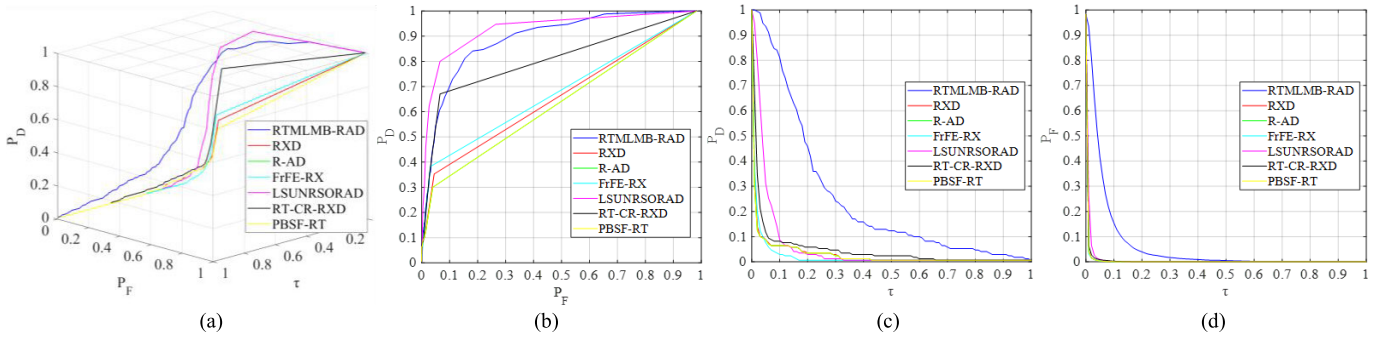


Fig. 20. ROC curves of different methods on Los Angeles-II: (a) 3-D ROC curves (P_D, P_F, τ); (b) corresponding 2-D ROC curves (P_D, P_F); (c) corresponding 2-D ROC curves (P_D, τ); and (d) corresponding 2-D ROC curves (P_F, τ).

no strong interferences in the scene. Compared with the local single-line information of the proposed algorithm, the LSUNRSORAD method makes use of local windows. By setting smaller windows, more local BKG distribution can be obtained. Hence, it has relatively high detection accuracy for anomalies in the Los Angeles-II dataset. It is worth noting

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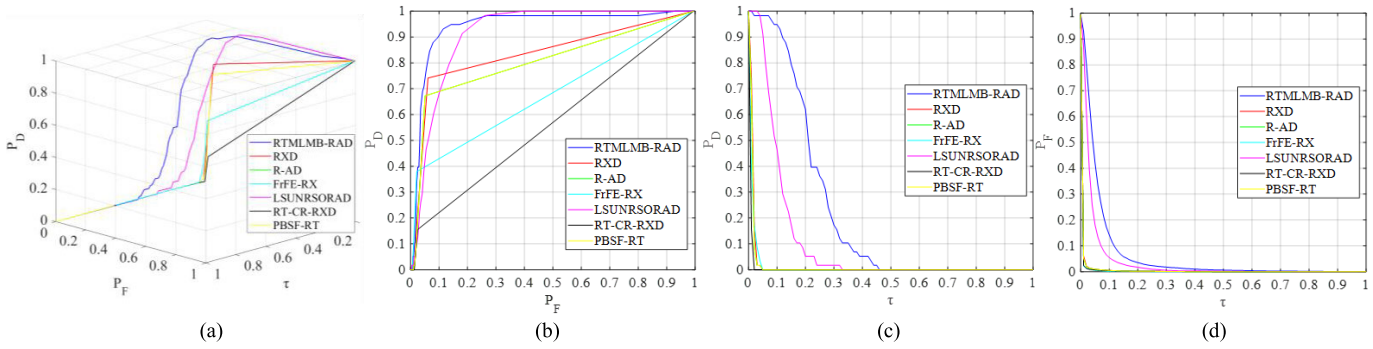


Fig. 21. ROC curves of different methods on San Diego: (a) 3-D ROC curves (P_D, P_F, τ); (b) corresponding 2-D ROC curves (P_D, P_F); (c) corresponding 2-D ROC curves (P_D, τ); and (d) corresponding 2-D ROC curves (P_F, τ).

TABLE VIII
AUC VALUES OF DIFFERENT METHODS ON LOS ANGELES-II

Method	AUC (P_D, P_F)	AUC (P_D, τ)	AUC (P_F, τ)
RTMLMB-RAD	**0.8737**	**0.2504**	0.0624
RXD	0.6435	0.0299	**0.0058**
R-AD	0.6199	0.0290	**0.0057**
FrFE-RX	0.6643	0.0290	0.0057
LSUNRSORAD	**0.9113**	**0.0563**	0.0090
RT-CR-RXD	0.7909	0.0472	0.0066
PBSF-RT	0.6199	0.0290	**0.0057**

TABLE IX
AUC VALUES OF DIFFERENT METHODS ON SAN DIEGO

Method	AUC (P_D, P_F)	AUC (P_D, τ)	AUC (P_F, τ)
RTMLMB-RAD	**0.9367**	**0.2245**	0.0615
RXD	0.8312	0.0145	0.0070
R-AD	0.8011	0.0129	0.0069
FrFE-RX	0.6738	0.0117	**0.0059**
LSUNRSORAD	**0.9093**	**0.1057**	0.0372
RT-CR-RXD	0.5609	0.0066	**0.0063**
PBSF-RT	0.8011	0.0129	0.0069

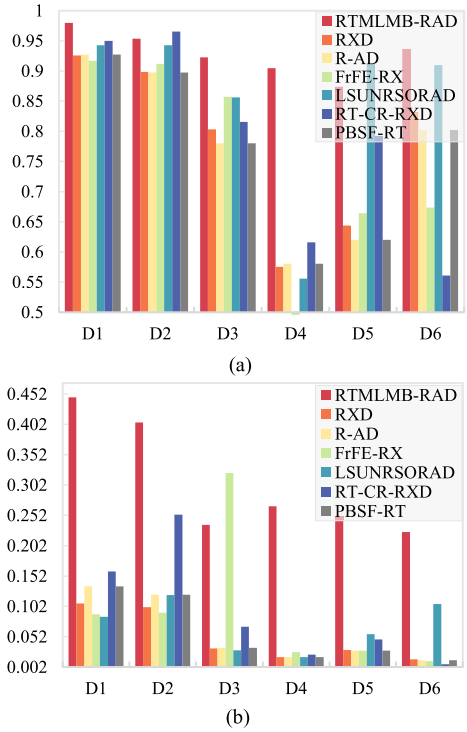


Fig. 22. Comparison of ROC values of different methods on six datasets: (a) AUC (P_D, P_F) and (b) AUC (P_D, τ).



Fig. 23. Industrial tobacco primary processing line.

that the AUC (P_D, τ) value of the proposed RTMLMB-RAD is the best compared with the comparison method, indicating that the proposed algorithm has a stronger retention ability for anomalies.

The ROC curve of each method on the San Diego dataset is shown in Fig. 21. According to the 3-D ROC curve and corresponding 2-D ROC curve in Fig. 21, the proposed RTMLMB-RAD has better comprehensive performance. As shown in Table IX, the AUC (P_D, P_F) value of the proposed RTMLMB-RAD is better than that of other comparison methods. This also verifies the efficiency of local anomaly detection methods although they perform better in different cases, respectively, because of the difference in local information. Significantly, the AUC (P_D, τ) value of the proposed RTMLMB-RAD is the optimal value compared with all comparison methods, indicating that the proposed method has a stronger retention ability for anomalies on the San Diego dataset.

In addition, to compare the performance of each method more intuitively, Fig. 22 shows the comparison of ROC histograms of different methods on six datasets. To fully display the data names, the Pavia Centre-I, Pavia Centre-II, Gulfport, Los Angeles-I, Los Angeles-II, and San Diego datasets are named D1, D2, D3, D4, D5, and D6, respectively.

It can be observed from Fig. 22 that the AUC (P_D, P_F) values of the proposed RTMLMB-RAD on the four real

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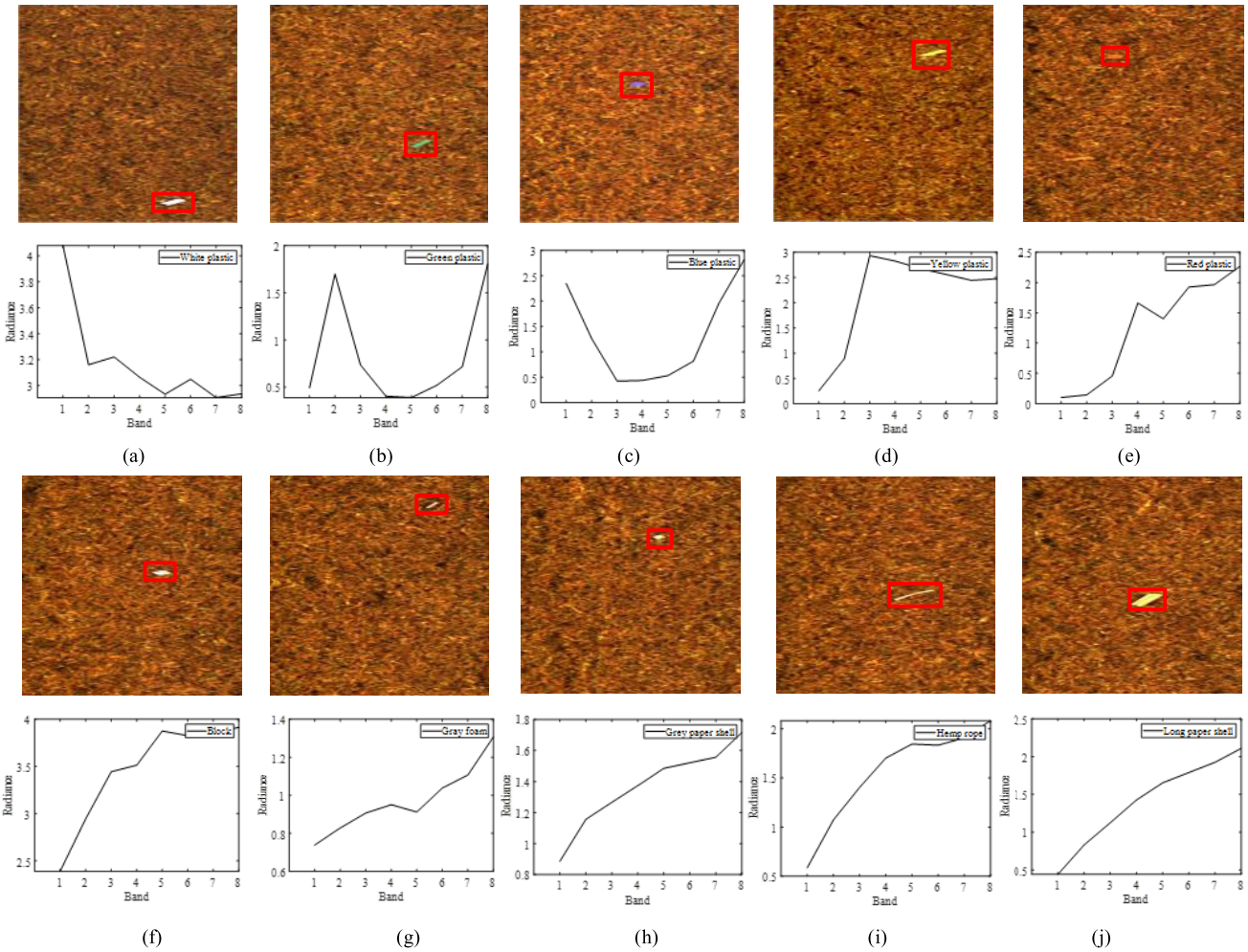


Fig. 24. Pseudocolor images, anomaly locations, and spectral signature of NTRM in real tobacco dataset: (a) white plastic; (b) green plastic; (c) blue plastic; (d) yellow plastic; (e) red plastic; (f) block; (g) gray foam; (h) gray paper shell; (i) hemp rope; and (j) long paper shell.

datasets are superior to all the comparison algorithms, and AUC (P_D, τ) values of the proposed RTMLMB-RAD on the five real datasets are superior to all the comparison algorithms, which indicates that the proposed method has better detection performance and stronger target retention ability. As mentioned above, although the AUC (P_F, τ) value of RTMLMB-RAD is not the best, considering that the better comparison method for BKG suppression will ignore some abnormal targets, the proposed method can detect almost all the number and complete abnormal targets. In general, the proposed RTMLMB-RAD algorithm shows a clear advantage over other methods in terms of anomaly detection performance in the widely used real hyperspectral data for anomaly detection.

C. Application of the Proposed RTMLMB-RAD on Real Industrial Tobacco Dataset

In this section, the validity and practicability of the proposed RTMLMB-RAD are verified by a real dataset captured from the tobacco primary processing line. There is a high-throughput lighting system in the scene. By comparing LED, halogen, and gold halogen light sources, it is found that LED has the highest light efficiency, halogen lamp has a higher brightness in the infrared spectrum, and gold halogen lamp can cover the visible light-short wave infrared spectrum with a higher light efficiency. Since the spectral characteristics of the tobacco and normal nontobacco-related materials (NTRMs) cover 450–900 nm, the gold halide lamp is selected for the hyperspectral identification of NTRM in the tobacco primary processing. Two gold halide lamps are used to illuminate the processing line on each side, supplementing light evenly as shown in Fig. 23.

Embedded in the system, the proposed method is tested online. According to the NTRM characteristics and tobacco, through band analysis and selection mechanism, the wavelength subset of eight bands selected from hyperspectral data is [488, 532, 586, 639, 694, 735, 849, 891], and the multiarray filter spectral camera is customized according to these wavelengths, whose spectral resolution is 10–15 nm. This article adopts the target-constrained interference-minimized band selection (TCIMBS) method [48] to determine the selected wavelength subset, which takes the spectra of various common impurities in the tobacco primary processing line as the target spectrum and uses tobacco as the BKG, to select multiple types of targets at once. Determine the minimum number of bands on the premise of achieving an acceptable industrial detection rate. Since the focus of this article is on anomaly

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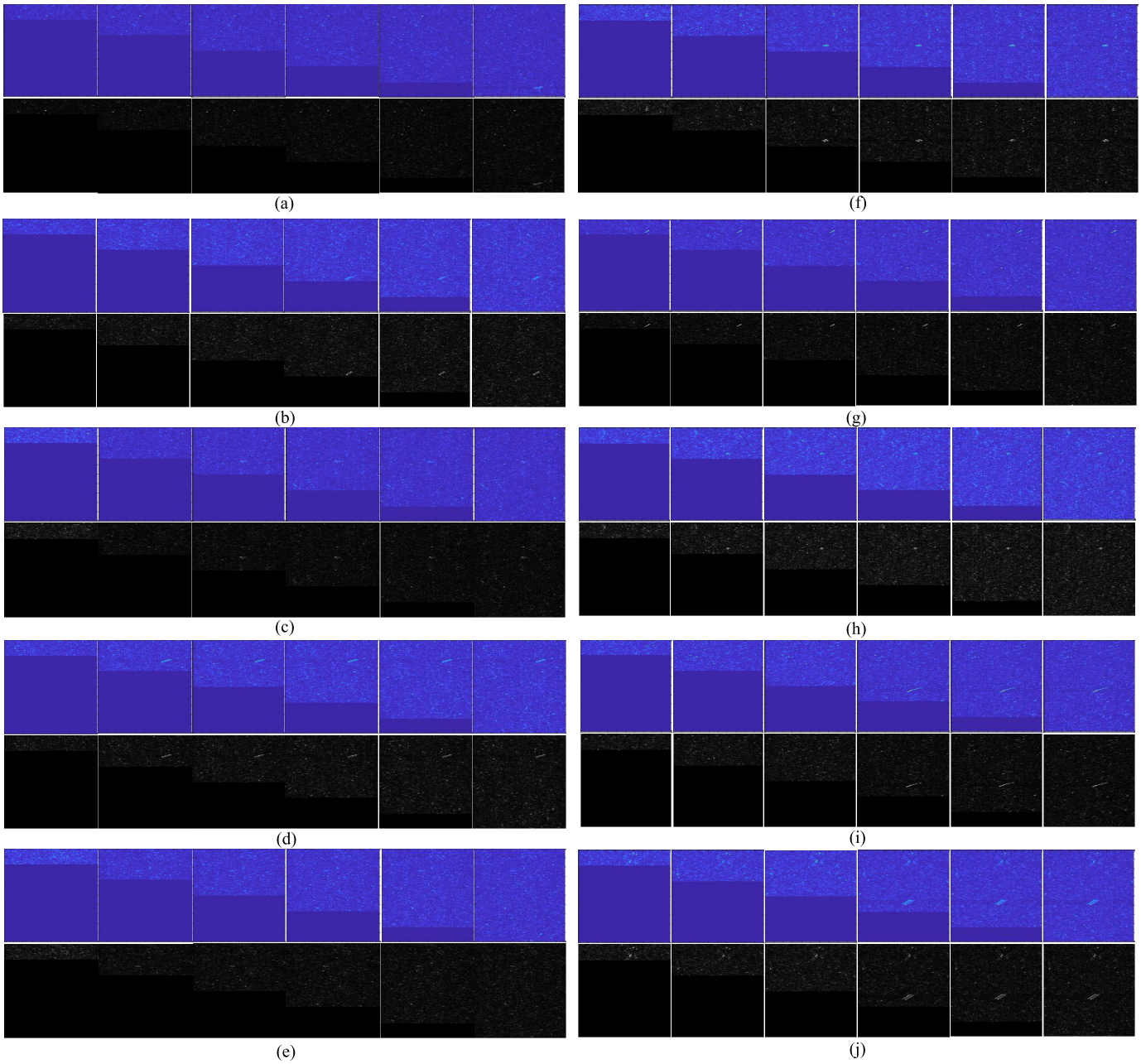


Fig. 25. Detection process of RTMLMB-RAD on real industrial tobacco dataset: (a) white plastic; (b) green plastic; (c) blue plastic; (d) yellow plastic; (e) red plastic; (f) block; (g) gray foam; (h) gray paper shell; (i) hemp rope; and (j) long paper shell.

detection, the content of band selection is not introduced in detail here, as well as the geometric correction and spectral dimension calibration. Ten images are picked out to illustrate the problem. The whole image contains 500 $\times$ 500 pixels, and its BKG is mainly tobacco. Ten types of NTRM including white plastic, green plastic, blue plastic, yellow plastic, red plastic, block, gray foam, gray paper shell, hemp rope, and long paper shell are regarded as anomalies to be detected. False-color images, anomaly positions, and spectral signatures of the target of ten large scenes are shown in Fig. 24, where the anomalies are circled in red boxes. This dataset is considered more challenging due to the fact that some tobacco-related materials, such as tobacco stem, would cause interference and be confused as NTRM and tobacco of different batches may change and bring different BKGs.

Fig. 25 shows the detection process of RTMLMB-RAD on ten real tobacco data of industrial scenarios, including color and grayscale. It can be observed that RTMLMB-RAD will detect weak false signals at the beginning, but when there is an anomaly target, the false signal will be ignored, and the suppression of the whole BKG will be gradually realized. At the same time, due to the interference of TRM in the image, there are false alarms in some BKG. In general, the proposed RTMLMB-RAD can accurately detect the location and shape of different sizes of anomalies in the tobacco scene and suppress most of the BKG in low detection value.

Table X shows the accuracy of the proposed method for the detection of various NTRMs. It can be seen that for the NTRM white plastic and long paper shell, the detection rate of the proposed method can reach 100%, indicating that the proposed

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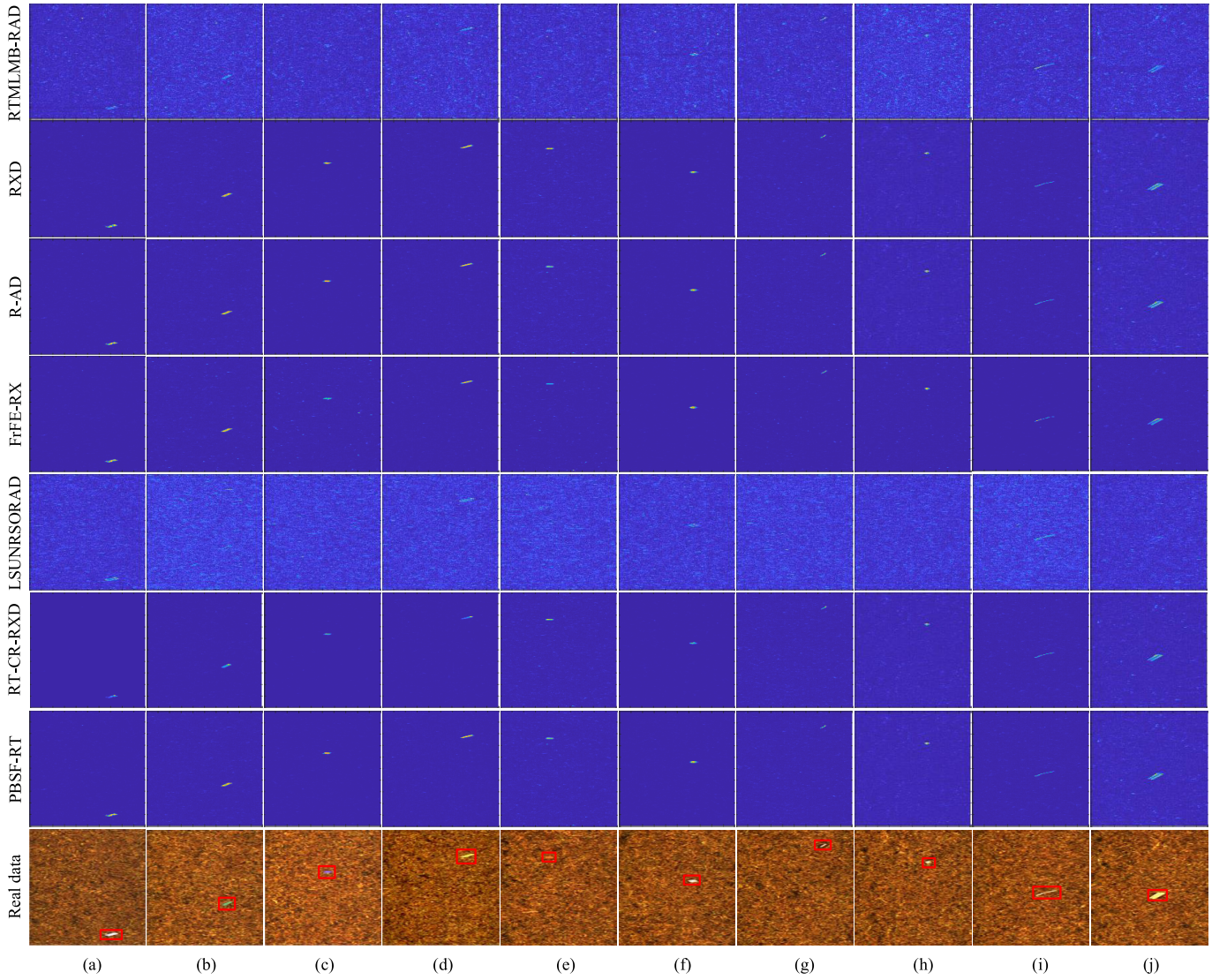


Fig. 26. Detection maps of different methods on tobacco data: (a) white plastic; (b) green plastic; (c) blue plastic; (d) yellow plastic; (e) red plastic; (f) block; (g) gray foam; (h) gray paper shell; (i) hemp rope; and (j) long paper shell.

TABLE X
RTMLMB-RAD METHOD FOR THE DETECTION PERFORMANCE OF TEN NTRMS

Type of impurity	Detection rate
White plastic	100%
Green plastic	92%
Blue plastic	82%
Yellow plastic	64%
Red plastic	93%
Block	83%
Gray foam	88%
Grey paper shell	93%
Hemp rope	82%
Long paper shell	100%
Average	87%

method has a good detection effect for these two materials. The detection rate of yellow plastic is lower because its signature is much similar to the tobacco. Meanwhile, for the ten NTRMs, the proposed method has an average detection rate of 87% and a false alarm rate of 10% at a speed of 3 m/s, which is acceptable for the industrial tobacco primary processing line.

The detection results of the proposed method and the comparison method on ten tobacco data are shown in Fig. 26. In order to better see the detection effect, the tobacco data abnormal position is circled with a red box in the last line of Fig. 26. As can be seen from Fig. 26, the proposed RTMLMB-RAD and the R-AD have almost the same detection results on the ten data sets, which proves that RTMLMB-RAD and R-AD have similar anomaly detection performance. For white plastic, block, and hemp rope datasets, the proposed RTMLMB-RAD, compared with the advanced FrFE-RX method, can achieve accurate detection of anomalies. In addition, compared with RXD, R-AD, RT-CR-RXD, and PBSF-RT, the anomalies detected by RTMLMB-RAD are more complete. This is because although the average energy of the anomaly target is not different from the global BKG sometimes, there will be a big difference in the local area. In this case, local anomaly detection will give

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excellent detection performance. It is worth noting that for the ten NTRMs, the proposed RTMLMB-RAD can identify complete targets compared with the local anomaly detection method LSUNRSORAD. As can be seen from the detection diagram, the proposed RTMLMB-RAD has obtained good detection results on all data due to the effective consideration of local information of BKG. In conclusion, the proposed RTMLMB-RAD algorithm can realize real-time online processing of MLMB data while ensuring detection performance, which provides a theoretical basis for the anomaly detection system of multiarray filter spectral cameras in the industry.

V. Conclusion

The low-cost multiarray filter imaging spectrometer is preferred for industrial scenario detection, which collects data by acquiring MLMB simultaneously. According to the principle of multiarray filter imaging spectrometer, this article prompts an idea of recursive updating based on MLMB and introduces local information into real-time online processing anomaly detection, proposing a real-time online processing local anomaly detection operator based on MLMB recursion to achieve real-time online processing detection of anomaly targets. It is worth noting that by introducing the block matrix to obtain the inverse formula, it is not only necessary to repeat the calculation of the existing information when acquiring each new data, but also to obtain the current detection results recursively through the MLMB update operator. During the operation of the algorithm, it only needs to store the information of the last time and the data of the current pixels to be proceeded, instead of storing the entire huge spectral data, especially suitable for applications where the decision cannot be delayed until the whole scene is obtained. The data storage space is greatly reduced, and the computation cost caused by complex matrix inversion operations is avoided. Experiments show that the proposed algorithm demonstrates stability and reliability under complex BKG, and can realize real-time data processing. Application of the algorithm to the real image from industrial tobacco primary processing line shows that it has great potential for practical industrial application. Although our focus in this work is on multiarray filter imaging spectrometer imaging system for industrial detection, the conclusions drawn from this work can be extrapolated to other domains that require real-time processing of hyperspectral images (e.g., spaceborne missions).

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